Bot Accounts

In Whose Name Does an AI Act?

Starting from the familiar device of the account, this book examines the identity and memory of AI, its work and delegation, its permissions and responsibility. It asks what must be identified, recorded and controlled if we are to entrust work to a particular AI and sustain that relationship over time.

Author: Ryo Ha

October 28, 2026

Contents

Resident Numbers, Account Numbers, Logins — and Bots 4

Part I **The Human World, Read through Accounts** 6

Ch. 1 How Names, Numbers and Accounts Differ 7

Ch. 2 How the State Registers People 14

Ch. 3 How Banks Record Money 16

Ch. 4 Who I Am after Logging In 18

Part II **The Moment an AI Needs an Account** 21

Ch. 5 From AI That Answers to AI That Acts 22

Ch. 6 Model, Bot, Account: How They Differ 24

Ch. 7 Is Yesterday's Bot the Same as Today's? 27

Ch. 8 The Birth and End of a Bot Account 29

Part III **How Memory Becomes Work** 31

Ch. 9 What Should a Bot Remember? 32

Ch. 10 Working Together through a Common Context 35

Ch. 11 Where Does Unfinished Work Remain? 38

Ch. 12 How Bots Recognise One Another 41

Part IV **In Whose Name, and How Far** 44

Ch. 13 Those Who Build, Operate and Commission a Bot 45

Ch. 14 Entrusting a Goal versus Granting Authority 47

Ch. 15 How to Prove What a Bot Did 51

Ch. 16 Where Responsibility for a Bot's Work Goes 54

Part V **Work and the Economy with Bot Accounts** 56

Ch. 17 Which Bots Will Individuals Keep at Their Side? 57

Ch. 18 How Will Organisations Take Bots In? 59

Ch. 19 When a Bot Spends Money 61

Ch. 20 Who Keeps the Door of a Bot Account? 64

Part VI **Questions for Designing a Society of Bot Accounts** 66

Ch. 21 Designing the Minimum Structure of a Bot Account 67

Ch. 22 If an Account Is Hijacked or Disappears 70

Ch. 23 Does an Account Create a Person? 73

Ch. 24 What Should We Entrust to AI? 75

To Grant an Account Is to Design a Relationship 78

Appendix 79

A. Key Terms 80

B. References and Notes 81

Resident Numbers, Account Numbers, Logins — and Bots

A resident registration number distinguishes me within administrative records. An account number points to a particular account at a bank. On the internet I enter through an account to write, to buy, and to form relationships with others. Set these different institutions side by side and a single question emerges. **How does a system recognise me, and how does it tie together what I have done and what I am permitted to do?**

One might at first call the resident registration number a “human account.” But to push that analogy to its end, the differences must be acknowledged first. A person's existence is larger than a registration number. A bank account carries a financial relationship; an online account carries rules of access and use set by the service. This book does not claim that these are one institution. It looks, in each distinct device, for the shared questions of identification, record-keeping and the management of relationships.

With AI the question becomes more concrete. When using a program that produces a sentence on request, the quality of the answer is what catches the eye first. But a bot that reads my files, carries on the work it was doing last week and reaches into other services needs an explanation beyond its answers. Which bot is acting now? Whose request is it carrying out? How far into my data may it look? If I stop it, do the requests it has not yet processed stop as well?

In this book, a bot account is a concept for managing these relationships. It is not completed by any one of the following alone: the ability to speak like a person, a distinctive name, or a long conversation history. **A criterion for identifying the bot must be set, and to that identity must be linked an operator, permissions, work and a record of activity.** Nor does this mean putting every function into a single database. It is a proposal to make the relationships scattered across many systems verifiable.

To carry the discussion forward, imagine a fictional publishing house. An editor, Jisu, assigns a research bot to collect material and an editing bot to review the structure of a manuscript. The two bots may run on the same language model. Yet the material each may read, the work each is assigned and the person who approves each one's results differ. This publisher and its people are a case constructed for explanation; they do not represent the operating results or findings of any real company.

Later chapters follow this publisher's bots as they revise their memory, receive a budget, change hands between staff and come under suspicion of account takeover. The work starts small, but the questions lead to organisations and to society. Who may issue and suspend an account? Can work be moved when a service is changed? From whom can a person harmed by an error demand an explanation?

Laws and standards tell us about distinctions that already exist. Research papers show possibilities and limits confirmed under particular conditions.

The structure of the bot account proposed by connecting those sources is the thesis of this book. It does not conclude that because something can be identified technically, consciousness or legal personhood has arisen. **Before we settle what an AI is inside, we must be able to explain within what relationships the work we entrust to it is carried out.**

PART I

The Human World, Read through Accounts

Chapter 1. How Names, Numbers and Accounts Differ

To understand accounts, we must first separate the names by which we are called, the identifiers that distinguish one thing from another, and the authenticators that confirm a right of access. An account is not quite the same as any of these. It is the unit of management a particular system uses to record a counterpart and to carry on its relationship with that counterpart. This distinction is the starting point for clarifying what it means to give an AI an account.

Suppose a company has two people with the same name. Both are Kim Jisu. When someone in a meeting room calls out “Jisu,” both may look up. Colleagues will tell them apart by adding a department or a job.

A system that pays salaries or sets access rights to project files, however, cannot be expected to guess from context each time. It must decide consistently which record connects to which person.

Suppose the company gives the two employees different staff numbers and creates work accounts for them. The names shown on screen are the same, but the staff numbers and account records are distinct. If one person's password changes or the other moves department, their work histories must not become entangled. Inside this ordinary case lie the differences between name, identifier, authentication and account.

1.1 The Name That Calls Me and the Number That Distinguishes Me

A name is used to call and to remember a person. Through a name we recall a face, remember shared experiences and continue a conversation with a particular person. In close relationships a name alone may suffice. But as people multiply and records accumulate across many places and times, we need rules for handling names that coincide or change.

What matters then is not the number itself but the scope of identification. That someone is staff number 42 in one company does not make them the same person as number 42 in another. “Number 42 of which company” must come with it. An identifier is interpreted within the system that issues and uses it.

NIST (the National Institute of Standards and Technology) is an agency under the United States Department of Commerce that sets standards for measurement and technology and publishes guidance on cybersecurity and digital identity. The digital identity guidelines, security control catalogues and AI risk management framework cited often in this book are all NIST publications. NIST defines a digital identity as an attribute or set of attributes that uniquely describes a subject within a particular context, and describes an identifier likewise as data that is linked to a single subject in a given context. The phrase to note is “particular context.” Knowing what a number points to in which system matters as much as knowing the number.[1] Nor should name and number be separated too rigidly. A name can serve as an identifier, and an identifier need not consist only of digits. A username a service does not allow to be duplicated, or a string that distinguishes accounts within a given scope, can play the role of an identifier. What we call an “ID” in everyday speech is an example. What matters is not whether the value is letters or digits but which object it distinguishes within which scope.

This difference can be seen in an actual specification. IETF RFC 7643, which defines the core schema of SCIM for exchanging user information, treats separately the displayName shown on screen, the userName that distinguishes users, and the id of the resource itself. The id in particular must remain stable and never be reassigned to another resource. This is an example of separating a human-readable name from the key a system uses to link records.[2] Applied to the company above, the two people may both appear as “Kim Jisu” on the HR screen while internally pointing to different objects. When an employee changes their name, managing the relation between the display information and the internal identifier allows their work history to continue. One cannot assume, however, that every service is implemented the same way. How name changes relate to account continuity depends on each system's rules.

Account numbers also help in understanding this distinction. ISO 13616-1 defines an account number as an identifier that identifies an account. What the account number points to is therefore the account. The slip of paper or string carrying the number is not itself the whole of the balance and the transaction relationship.[3] An identifier need not explain everything about a person. The string that distinguishes a work account need not contain the person's tastes or family relations. The system only needs to be able to link the records it requires. This is where “telling who someone is” diverges from “knowing everything about that person.”

This distinction raises the same questions for bots. When the name “ResearchBot” appears twice on screen, the name alone cannot tell us whether one bot is running in two places or two different bots share a name. We must check which identity the name is linked to and how the system distinguishes the two. Naming a bot and managing a bot's identity each require their own design.

Applying the distinctions so far to four institutions side by side makes the limits of the “account” analogy clear. A number is a clue for finding which object is meant; records and permissions are managed by separate rules. A number alone creates neither access rights nor legal personhood. Carried into implementation, the display name is separated from the internal ID, and the secret keys and tokens used for authentication are kept apart from the account information. For example, if a bot's display name is changed to “ResearchBot,” the earlier work records of the internal ID bot-01 must still be retrievable.

Domain	Value that distinguishes the object	Linked records and relationships	Conditions that decide access and action
State registration	Resident registration number	Resident register and other administrative records	Relevant law and administrative procedure
Bank account	Account number	Holder, balance, transactions, contractual relationship	Transaction authority and the institution's procedures
Internet service	User ID within the service	Profile, activity, settings, permissions	Authenticator + the service's access policy
Bot operating system	Bot account ID	Operator, work, runs, memory references	Bot identity check + per-action permission check

〈Table 1〉 Identifier, records and access conditions in four institutions — this does not imply that their legal status is the same

1.2 Identifier, Authenticator and Account

At the login screen these differences are hard to see. Enter a username, enter a password, and the service opens at once. Because the two fields are always used together, it is easy to think of the username–password pair as the account itself. Behind that screen, however, lie different questions. Which record is being accessed, is the authenticator linked to that record under the requester's control, and is the requested operation permitted — each must be judged separately.

NIST's digital identity model describes a subscriber account as the record an identity service provider maintains about a subscriber, their attributes and the authenticators linked to them. In this description an account is broader than a line of username or a single password. An authenticator is an element linked to the account, and a verifier checks that link and the state of the account.[4] An identifier tells us which object is meant. Authentication is the process of confirming that the party claiming an identity possesses and controls a registered authenticator. A password is one secret that can be used for this; NIST treats it as an authentication factor of the “something you know” kind.[5] Knowing another person's username must not let you into their account. The same distinction appears in software. RFC 6749, the base specification of OAuth 2.0, states that the client_id identifying a registered application is not a secret and must not be used on its own to authenticate the client. **Knowing a name tag and confirming the right to act as the party the tag points to are different procedures.**[6] The result of authentication must also be interpreted within its scope. That a registered authenticator was used does not prove that the legitimate user of the account personally intended the action at that moment. The difference is plain if the authenticator has passed into someone else's hands. Authentication must therefore be understood together with what evidence was checked and how far that evidence extends trust.

After authentication, permissions must be checked. This process is called authorisation. Even if two employees on the same project both log in normally, one may be allowed only to read a document while the other may edit it. **That an identity has been confirmed and that a particular action is permitted are separate judgements.**

NIST's guidance on attribute-based access control describes deciding whether to allow access by comparing the subject, the object, the requested operation and, where needed, environmental conditions against policy. From this viewpoint, access rights do not end with a single “allowed” flag attached to an account; what is being accessed and what is being attempted are weighed together.[7] Suppose the company replaces an employee's password. The information used for authentication changes, but the system can be designed so that the employee's work account and project history continue unchanged. Conversely, only the document-editing right may be withdrawn from the same account. Changing a display name, changing an authenticator and changing permissions are each different events.

Missing this difference makes it hard to describe problems that arise in operation. What needs fixing differs according to whether a login failed, a login succeeded but access to a document was refused, or data was linked to the wrong account. The first may be an authentication problem, the second an authorisation problem, and the third a problem of linking identity and records. To the user they all appear as “I can't get in” or “my files are gone,” but the point to check inside the system differs.

That records are linked to an account does not mean all related data must be stored in one place within the account data. One can imagine a structure in which the account management system, the document store, the permission policy and the activity log are kept separately and linked. The role of the account described here lies not in a physical storage location but in providing a consistent basis for managing relationships.

〈Figure 1〉 A name tag points to an object; a key confirms a right of access. To an account are joined the relationships of records and permissions.

1.3 Having a Place within a System

Suppose a work account has been created at the company. Now the person can be registered as a participant in a particular project, documents can be shared with them and work assigned. When they connect again the next day, the same records can be found and linked. An account removes the need to explain who the counterpart is from scratch each time and provides the basis for carrying earlier relationships into present work.

The picture of a person that a system holds is assembled for a particular purpose. In a company's work account, department and role may matter; in a library lending account, loan status and due dates. That one person holds several accounts does not mean the person is divided into several beings. It means, rather, that different systems form relationships with the same person according to their own purposes.

NIST's subscriber account guidance also covers a person acting in several roles or personas — distinguishing, for example, acting as an individual from representing a business. The guidance separates the handling of such roles from the handling of multiple accounts where these are allowed; it does not conclude that every role requires a new account.[8] Nor does an account always demand a legal real name. The same guidance explains that a pseudonymous account may be created for an applicant who has not undergone identity proofing, with that status recorded. That an account consistently points to the same counterpart and that the counterpart's real-world identity has been verified can be distinguished.[8] Think of someone active under a pen name in a reading group. Other members can connect that person's earlier and later writing to the same account without knowing their legal name. This is one possible service design. That the account's writing can be grouped as one person's does not entail that everything about the person behind the pen name must be disclosed.

This is why an account can be called “a place within a system.” That place is not a vessel holding the whole of a person's existence; it is the way a system decides which counterpart to connect to which records and rules. Linking records wrongly can show someone else's documents or assign the wrong task. Connecting relationships accurately becomes a central task of account management.

When an account's state changes, the conditions under which the relationship can be used change too. An account moved to dormant or suspended status can be operated so that it cannot be accessed as before. NIST guidance treats separately the renewal, suspension and termination of accounts and the retention and disposal requirements after termination. The act of terminating an account is not the same as a state in which all related records vanish at once.[8] Here lies an institutional dimension that technology alone cannot explain. Which records are accepted as valid, to whom data is shown and under what conditions access is withdrawn are governed by rules set by the operating body. Technology implements those rules, but it does not decide on its own what to permit.

The philosopher John Searle, explaining institutional status, argues that functions which cannot be established by an object's physical properties alone operate within social recognition, and that such status is linked to relationships of rights, duties and permissions. This is one theory to consult when thinking of an account as a combination of recorded information and recognised relationships. Searle's argument does not, however, directly prove anything about digital accounts or the personhood of AI; the connection to accounts drawn in this chapter is the author's interpretation.[9] On the basis of these distinctions, this book understands an account as follows. **An account is the unit by which a particular system distinguishes a counterpart, links the information and access relationships concerning that counterpart, and manages the state of that relationship.** This definition is drawn up for the purposes of this book on the basis of digital identity management concepts. It does not mean that the legal and technical structures of administrative registration, bank accounts and online accounts are all the same.

Let us carry the questions we asked of people over to software. If a program reads a document and saves a result, the system must decide whose request it treats this as. W3C's decentralised identifier specification, DID Core, also explains that the subject identified is not limited to people and may be an organisation, a thing or a digital object. **Being able to be the subject of an identification scheme and having personhood are separate matters.**[10] Giving a bot an account can start from this distinction. If a bot has been named, people can call it. If it has been given an identifier, the system can tell it apart from other objects. Add authentication, permissions and related records, and the conditions under which that bot operates can be managed. Only when it is clear how far implementation has gone does the phrase “bot account” describe an actual function.

Suppose there is a document produced by the research bot. Writing the bot's name on the document is not enough to tell in which run it was produced, which material the bot was permitted to read, or at whose request it worked. Linking work and permission records to the account's identity gives a starting point for answering these questions — though how to build and verify that link remains a separate design task.

That relationship is what this book sets out to explore. When we say we have entrusted work to a certain AI, what do we recognise as the same counterpart, and which actions do we leave as that counterpart's record? To address the question, we first need to look at the relationships that registration and accounts have built in the human world. The next chapter turns to how the state registers people.

Chapter 2. How the State Registers People

2.1 Linking the Resident Registration Number to Administrative Records

Suppose someone has moved house and changed their address. Their body has already arrived at the new home, yet the administrative record may still hold the old address. A person's move and the updating of a record are not the same event. A registration system carries changes that occur in reality into records through a fixed procedure. Thinking about this time lag reveals the role of a registration number. The number does not create a life, but it is the basis for distinguishing whose record is being handled across many administrative processes.

Article 7 of Korea's Resident Registration Act governs the creation and management of individual and household resident registers, and Article 7-2 provides that each resident be assigned an individually unique registration number. It is worth noting that record and number are provided for in separate articles. The number is not the same as the record as a whole, nor is the system that manages the record explained by the number alone.[11] Calling the resident registration number a “human account” can therefore serve as an analogy for thinking about structure. Strictly, it is the relation between an administrative identifier and the registration system that uses it. When comparing it with bank or online accounts, purpose, who is registered, correction procedures and access conditions must each be examined. That similar-looking numbers exist does not make them the same institution.

〈Figure 2〉 A person moving and the address in the administrative record changing are different events. A procedure for correcting the record is needed.

2.2 A Person's Existence and Their Registered Identity

Registration is important for handling social relationships, but human rights must not be explained as beginning with a registration number.

Article 3 of the Korean Civil Act provides that a person is the subject of rights and duties throughout life. The wording does not define legal capacity by whether an account has been issued.[12] The limits of the comparison become clear where the registration procedure is distinguished from the status of being a subject of rights.

The United Nations legal identity agenda likewise treats together the right to recognition as a person before the law and the importance of birth registration. Here registration is a means of officially proving an individual and connecting them to institutions.[13] The fact that a person lacking documents may have difficulty using institutions must not be confused with the fact that the person holds rights as a human being.

This distinction is useful when registering bots too. Giving a program a number means the operating system has decided to distinguish it. It does not follow that the program has thereby become a subject of rights like a human. What we borrow from human registration systems is the set of questions about linking and correcting records, not an analogy that copies human status wholesale.

2.3 People Who Are Unregistered or Wrongly Recorded

UNICEF explains that birth registration and certificates are important for confirming a child's identity and for access to essential services. Gaps in registration can make connection to health, education and protection systems difficult.[14] Because systems differ by country, this cannot be equated with the effects of Korean resident registration. It does show, however, that being left out of a record system can go beyond a mere paperwork inconvenience.

A record that exists can still be wrong. If, say, the information of two people with the same name is wrongly linked or a change of address is not reflected, the system's very consistency becomes a means of repeating the error. Hence a procedure for correcting wrong records is needed. Article 21 of the Resident Registration Act provides for objections to certain dispositions such as correction, deletion and registration of unknown residence.[11] The efficiency of a record must be weighed together with the possibility of correcting it.

The operating principle this book draws here is clear. **A bot account, too, must allow wrongly linked records to be challenged and corrected.** In particular, if a person is denied access or wrongly assessed because of such a record, the answer “that is what the system says” must not be the end of it. Registration stabilises relationships, but it becomes trustworthy only when it also provides a channel for putting those relationships right.

Chapter 3. How Banks Record Money

3.1 The Financial Relationship an Account Number Points To

The numbers on a bank statement are of two kinds. One distinguishes the account; the other is the balance. Knowing the number does not let you alter the balance at will, and an equal balance does not make two accounts the same account. A bank must handle together the identification of accounts, the recording of financial transactions and the authority of the person requesting a transaction.

The Bank of England, explaining modern money, treats bank deposits as a form of liability owed by the bank to households and firms.[15] This description makes it hard to equate the sum on a statement with a personal name tag stuck inside a vault. A balance is a record representing the relationship between a financial institution and its customer. The specific contracts and rights involved vary by country and product.

Suppose the fictional publisher has operating funds in its account. The director may check the account and the accounts clerk may handle transfers. That both see the same balance on screen does not make the ownership of the money and their respective powers the same. To describe an account one must know not only which number it is but whose account it is and what transaction relationship it carries.

〈Figure 3〉 An account number points to an account. The current balance must be read together with the transaction history, and ownership of the money must be distinguished from the authority to move it.

3.2 Rights and Duties in the Balance and the Transaction History

Keep only the balance and discard the transaction history, and it becomes hard to explain how the present sum was reached. A printing payment, a royalty receipt and a refund for a cancelled order each have different grounds. The moment an account is used for business, its sums must be linked to other records — invoices and approvals, contracts, transfer results. This is an example for understanding the fictional publisher's bookkeeping.

Article 3 of Korea's Act on Real Name Financial Transactions requires financial companies to conduct transactions under the transactor's real name and presumes that the financial assets in certain real-name-verified accounts belong to the named holder.[16] The word “presumes” also deserves care. The name on an account screen must not be read as settling final ownership in every dispute. Judging the legal relationship requires the applicable law and the specific facts.

This distinction can be applied to a bot's resource management. Allocating a material-purchase budget to the research bot does not immediately mean the bot has acquired property of its own. A budget may be a limit the organisation has set for use toward a particular purpose.

One must distinguish whether a figure shown on an account is ownership, an available limit, or an amount already spent.

3.3 Those Who Hold Money and Those with Authority to Move It

Suppose the publisher has the research bot buy a paid journal article. The bot may propose the price and the item, the editor may approve, and a payment service may process it with the company's means of payment. Here lie several distinct functions: the purchase judgement, the approval, the execution of payment and the bearing of the cost. This is not a description of what any particular financial service already provides but an example for designing operations.

As work grows, a single “may pay” permission becomes vague. The authority to buy one article differs from the authority to start a monthly subscription. It must also be decided whether a once-approved price needs re-approval when it changes. Simply telling the bot the payment details cannot express such differences adequately.

The insight from bank accounts is that the technique of recording money is also a technique of handling relationships. In a design that entrusts resources to a bot, showing only a balance or a limit is not enough. It must extend to whose purpose the resource serves, under what conditions it stops, and by what record misuse can be confirmed. Chapter 19 returns to this problem as a concrete flow of budget and payment.

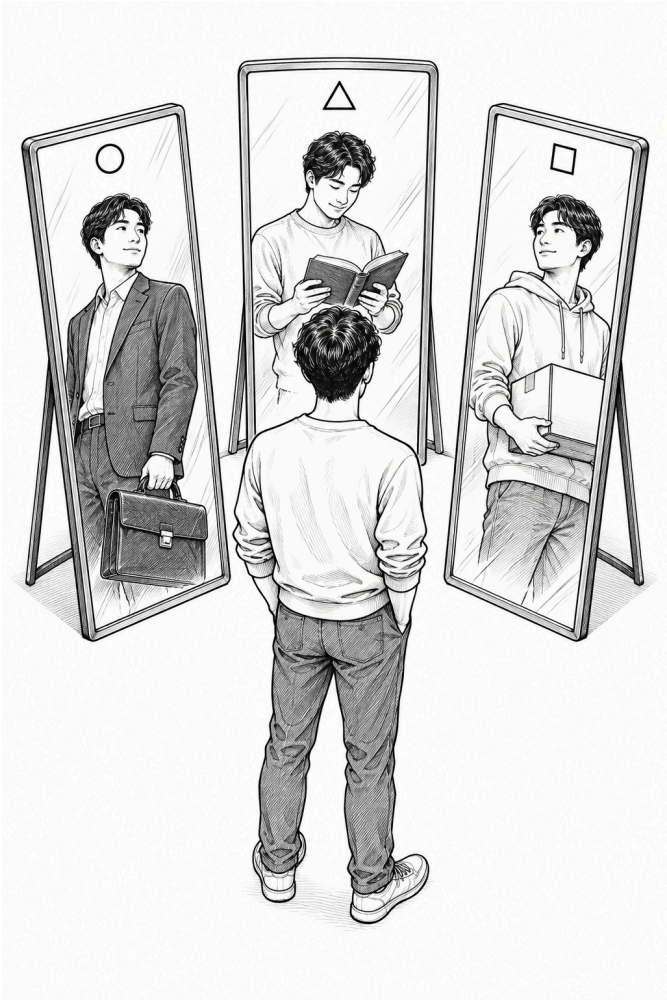
Chapter 4. Who I Am after Logging In

4.1 One Person, Many Internet Accounts

Suppose one person is active under different names in a work email, a reading group and an online shop. The work account is linked to a role in the organisation, the reading group to writing and conversation, the shop to orders and delivery addresses. The actual person is one, but the relationships each service has with that person differ. Holding several accounts is a way of separating those relationships according to purpose.

There is also federated identity, in which several services are used through one authentication procedure. NIST's federation guidance describes a structure in which an identity provider conveys verifiable assertions about authentication to a service.[17] Sharing a login procedure does not mean merging all services' accounts and stored data into one. The rules that connect authentication and the rules by which a service manages usage records can be kept apart.

For instance, even if company authentication admits someone to the archive, the archive's record of what was viewed may be managed in a system separate from the mailbox. This distinction becomes important later when a bot is connected to several tools. It must not be assumed that, because something has been authenticated once, the contents of every store and every operating permission naturally follow.



〈Figure 4〉 A person using several accounts is not thereby divided. What differ are the records and access scope of each service.

4.2 From Passwords to Authentication and Access Rights

An account cannot be understood around passwords alone. W3C's WebAuthn Level 2 specifies a method of web authentication using public-key credentials; a service can authenticate through verification related to a registered public key.[18] What this case shows is that the continuity of an account is not the same as the survival of a particular password string.

Even when the method of authentication changes, the problem remains that a service must decide whose request to connect to which account. Procedures are also needed for returning to existing records when a phone is replaced or an authentication device is lost. Because convenient recovery and the exclusion of others must be achieved at the same time, the recovery process can reveal who really controls an account better than the initial sign-up screen does.

Access rights are yet another layer. That I can edit my own documents in the company archive does not mean I can delete another department's. NIST's zero trust account does not grant trust automatically for being inside the network or for owning an asset, and it distinguishes authentication from authorisation.[42] That a bot runs inside the company likewise cannot substitute for per-task permission judgements.

4.3 Activity, Relationships and Reputation That Accumulate on an Account

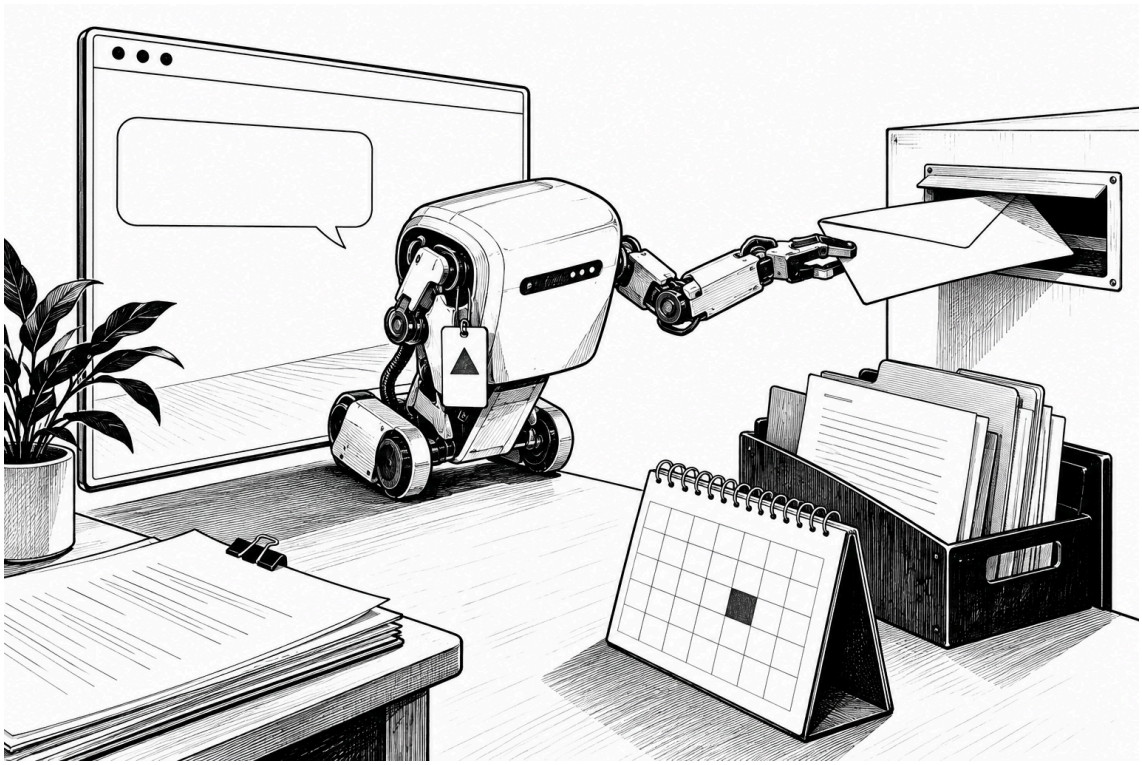
Over time an account becomes the basis on which past behaviour is gathered. Posts in the reading group, purchase reviews at a shop and the revision history of a shared document can influence the next interaction. But a high volume of activity and the accuracy of the record are separate matters. If someone shared the account or an automation tool performed some of the work, the record cannot be read straight off as an individual's intent and ability.

Suppose the fictional publisher runs all its bots under editor Jisu's personal account. Only Jisu appears in the record of document edits, yet in reality the research bot and the editing bot may have worked at different stages. To find the cause of an error, which run did what must be checked separately. Attaching automation to a person's account is not by itself enough to distinguish a bot's actions.

The questions that lead from internet accounts to bot accounts therefore begin inside the familiar login screen. To whom does this request connect? As whose action can the record be interpreted? May earlier trust be applied to the next task? As what bots can do in the outside world grows, such questions move beyond technical details behind the access screen and become basic conditions of work.

PART II

The Moment an AI Needs an Account



Chapter 5. From AI That Answers to AI That Acts

5.1 The Distance between Generating an Answer and Performing Work

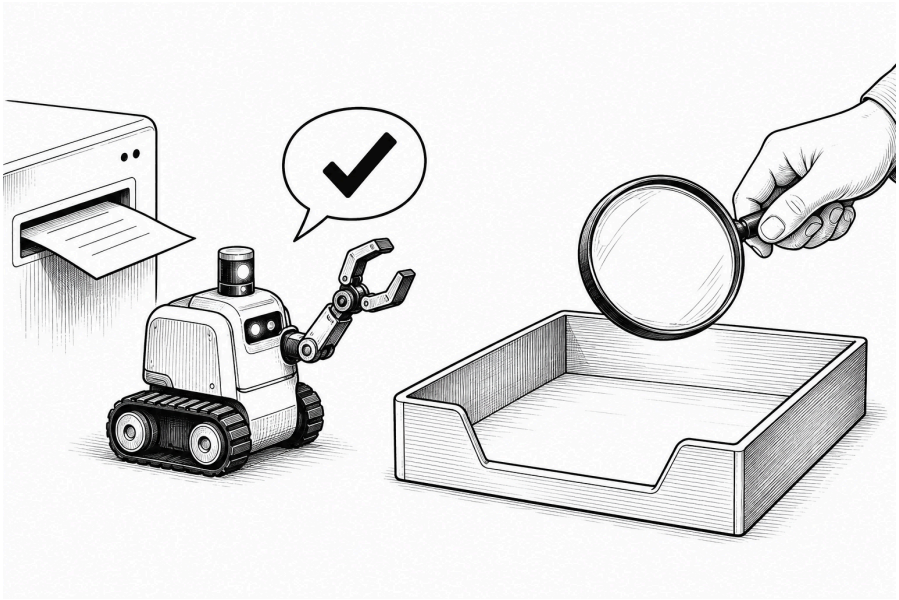
Jisu asks the research bot to find references on a topic. If the bot stops at suggesting search terms, Jisu will look up the material herself. If the bot uses a search tool, it can bring back a list of results; if access to the archive is allowed, it can read the full texts. If purchase and correspondence are possible too, costs may be incurred or messages may reach people outside.

For the same request, what actually changes depends on the functions connected.

The ReAct study proposed a method in which a language model alternately generates reasoning about a task and actions, and feeds information obtained from the outside environment into the next step.[19] This is not a result guaranteeing success in every task. But it shows concretely a structure that links language generation to tool use. The moment an answer becomes input to another program, the sentence and the execution must be evaluated together.

The bot saying “material secured” and a file being saved in the archive are events that can be confirmed separately.

An account of having attempted a purchase and the fact of a completed payment also differ. A work system should take the bot's report into account but confirm the actual state of external systems. If the fluency of language substitutes for confirming results, a failed task may be recorded as complete.



〈Figure 5〉 A bot's report of completion and the actual result left in an external system must be confirmed separately.

5.2 Bots That Use Tools and Reach External Systems

Tools determine not only a bot's capability but its sphere of effect. A search tool and a file-deletion tool differ in how reversible their results are. Even with the same model, different tools may call for different supervision and recovery. In this book the unit for evaluating an acting bot is not the model's answer score alone but the work structure that binds model, tools, permissions and operating procedure together.

For example, when the research bot sends a request to read a document from the archive, the archive must know which identity to connect the request to. One could implement a scheme that unconditionally hands over all of a person's permissions, but that may allow access excessive for the purpose of the work. Limiting the research scope to a single project folder keeps other manuscripts and contracts out of the task. This is a design chosen to fit the present case.

Reading external material is not entirely passive either. OWASP describes the risk of indirect prompt injection, in which the content of a web page or file steers a model's behaviour in unintended directions.[25] If the research bot treats instructions inside a document it reads as the user's commands, the channel opened to obtain information becomes a channel for misusing permissions. The problem of separating material to read from instructions to follow leads on to the later design of the common context.

5.3 When the Agent of an Action Must Be Distinguished

Not every text-generating program needs its own long-term account. A function that performs a single calculation or summary can be managed as part of an existing work service. What matters is at which level actions must be distinguished. As several requesters, long-running work and independent permissions appear, the unit of distinction becomes finer.

If the publisher runs the research bot for a month, it must distinguish who revised yesterday's result from which request triggered today's purchase. An account can be a unit of responsibility maintained over a long period, and a run identifier can distinguish individual tasks within it. Recording account and run together narrows “the research bot did it” to a specific event.

The need for a bot account does not depend on how human the bot appears. It depends on whether external systems need to distinguish and control its actions. A short program without a name or a face may need strict permission management, while a program that chats warmly may have no authority to act externally at all. The design of an account must start from the actual work, not from the manner of expression.

Chapter 6. Model, Bot, Account: How They Differ

6.1 The Model That Provides Capability and the Agent That Performs Work

Suppose a publisher uses the same model for research and for proofreading. For research it connects a search tool and a reference store; for proofreading, the manuscript and the house style rules. Though the model is the same, the material read and the actions available differ, so the behaviour of the two tasks may differ. This is why a model name alone cannot explain what a given bot can do.

In this book, a model is the core component that processes input and generates output, and a bot is the software configuration built to perform work by combining that model or another program with instructions, tools and state. The term agent is used when the focus is on the function of choosing the next action and continuing the task on receiving external results. These are distinctions for the sake of discussion; they do not mean the industry has agreed on a single definition.

A bot account is the unit that manages the identity and operating relationships connected with this configuration. A model version may be linked in the account record, but **the model itself is not the account.** Conversely, replacing the model does not necessarily require creating a new account. What is kept and what is newly identified must be set by the service's continuity rules.

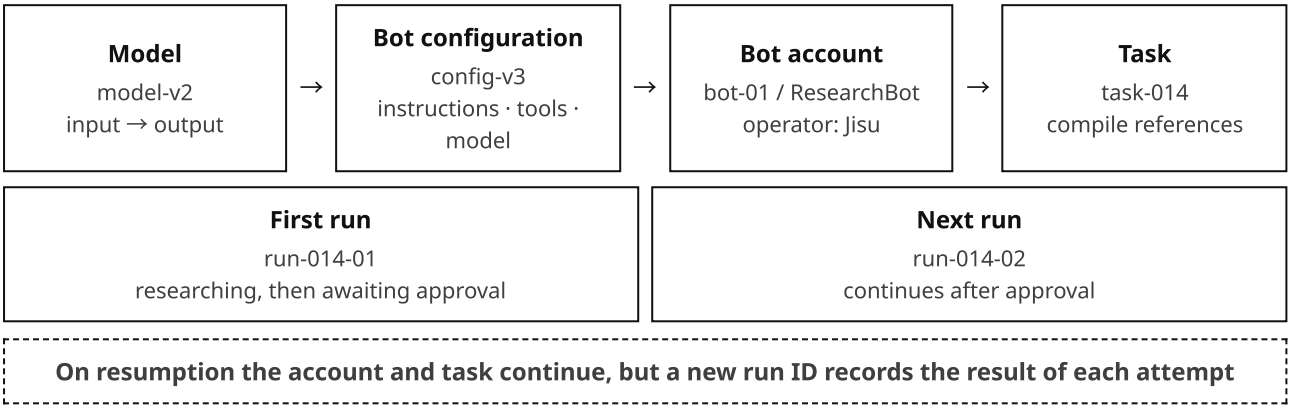
6.2 The Account's Identity and Individual Runs

Suppose the research bot account investigates historical material in the morning and scientific material in the afternoon. The account is the same, but the requester, the material used and the results produced differ. The two tasks must therefore be recorded as separate runs if the provenance of each is to be followed. Even within a single task, retries and tool calls can be recorded as smaller events.

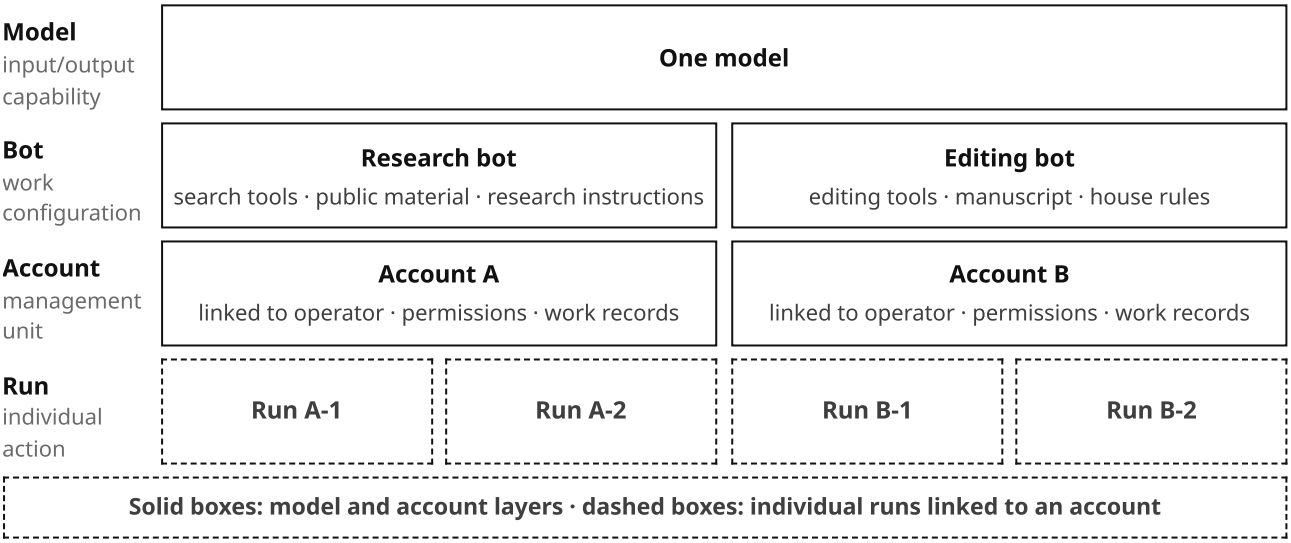
SPIFFE, which deals with software identity, explains that a workload may consist of several running instances, and distinguishes the workload identifier from the document that attests to it.[20] This case shows that a logical unit of work need not always map one-to-one to a single running program. It does not mean, however, that adopting SPIFFE alone manages a bot's memory, work and responsibility.

At the publisher, one research bot account may be shared by several runs. If only the shared account is kept and the run records discarded, tracing the cause of concurrent work becomes difficult. Linking an account identifier, a task identifier and a run identifier is not a matter of multiplying names; it is a matter of answering different questions. Which responsible bot? What work was it doing? In which run did it happen?

Drawn in one picture, the relationship is as follows. The model is a component of capability, the account is the unit of responsibility, and the run is a single action that actually took place. In implementation, account_id, task_id and run_id are kept separately, and the model and tool configuration version used in each run is recorded. **Replacing the model does not mean creating a new account**, and whether two runs hold the same permissions must be checked again.



〈Figure 6〉 The relationship between model, bot configuration, account, task and run



〈Figure 7〉 Different bots can be built from the same model, and several tasks and runs can follow under a single account.

6.3 Different Bots Using the Same Model

That the research bot and the editing bot run on the same model is no ground for letting both see the same material.

It can be decided that the research bot accesses public material while the editing bot reads unpublished manuscripts. Even when the two cooperate, the editing bot's whole manuscript need not be passed to the research bot. A structure that passes only the necessary questions and permitted excerpts can be chosen.

Conversely, one bot may be configured to use several models — a small model for classifying documents, another for reviewing long manuscripts. In that case the record of which model was used differs, but the work promised to the user and the operating responsibility can continue under the same account. Here the account is better understood as the name of a service relationship than as the name of a model.

This distinction also helps when problems arise. One can investigate separately whether an error in an answer was due to the model's limits, to wrong material being passed in, or to a tool holding excessive permissions. Lumping every problem together as “the AI was wrong” makes it hard to find what must be changed. Account design is the work of connecting these components while keeping them distinguishable.

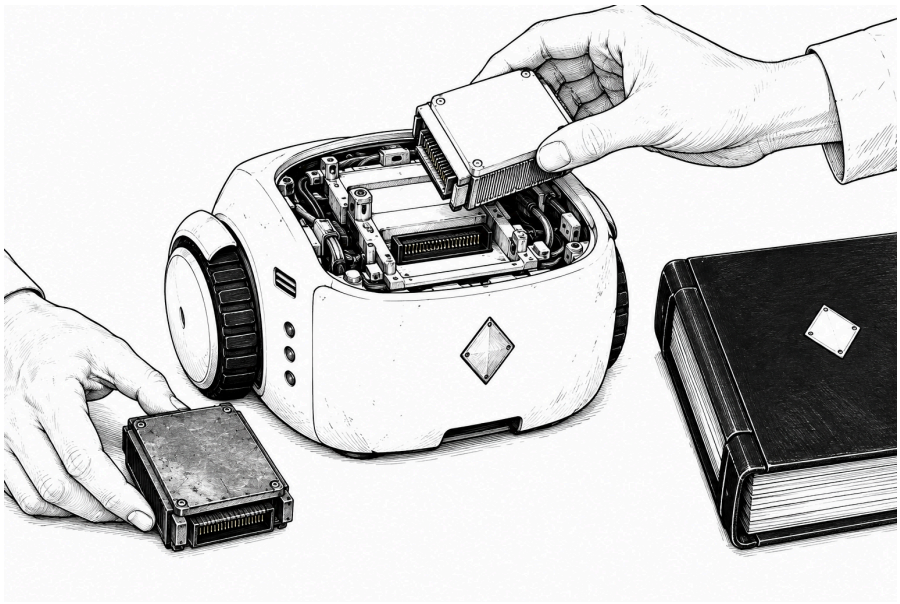
Chapter 7. Is Yesterday's Bot the Same as Today's?

7.1 What Continues When the Model and the Memory Change

Suppose the model behind the research bot used until yesterday is replaced today. The account name is unchanged and last week's material and unfinished tasks are still accessible. Yet the style of its answers and its choice of sources may differ. The user feels they are working with the same responsible bot while experiencing a change in results. Here what “the same” means must be made concrete.

At least three things can be distinguished: operational identity, similarity of behaviour and continuity of stored state. That an account continues means the operating system manages it as the same unit of responsibility. It guarantees neither the same answers nor the preservation of every previous state. Only when the fact of the model change and the results of review after it are recorded can continuity and change be explained together.

The SCIM requirement for a stable identifier seen in Chapter 1 is a technical example of keeping a basis for distinguishing a resource even when its display information changes.[2] A similar idea can be applied to bot accounts, but one must not claim that because the identifier has not changed, the capability has not changed either. Just as the chef at a restaurant with the same address can change, the rule that carries identity forward and the procedure that evaluates performance are each needed.



〈Figure 8〉 The account and work records can continue while the model changes. Being the same account does not guarantee the same performance.

7.2 Identical Copies and Separate Accounts

Suppose the research bot's configuration and memory are copied to create a new bot in another department. Just after creation the two may answer similarly. But if one is put to editing general-interest books and the other to specialist titles, their later work and approvers diverge. That the initial data was the same is no reason to treat the two operating relationships as one.

In this case a design can be proposed that grants a new account and records the relation of having branched from the original. The copied memory, too, carries its source and the time of copying. It must also be decided whether a correction to a source error found on one side is reflected automatically on the other or only after review. Copying may look technically simple, but copying the working relationship is a separate decision.

If two copies share the same authenticator, this distinction blurs, because external systems may see only one identity. This does not mean every parallel run must be given a new account. But once a branch arises with a different operator and purpose, it must be clear whether an identical run has been multiplied or an independent responsible bot has been created.

7.3 Technical Continuity and Philosophical Identity

That yesterday's conversation was loaded today shows continuity of storage and retrieval. It does not by itself tell us whether any subjective experience continued. Conversely, that part of the memory was deleted does not mean the operating account must be closed. Questions that can be answered in technical management must be distinguished from questions about consciousness.

In practical work it is more useful to set verifiable criteria first. If the same account continues, check whether the operator is retained, who takes over unfulfilled commitments and whether earlier permissions have been reviewed. If branching to a new account, state how far earlier reputation and memory are carried over. These criteria can be designed without a metaphysical answer.

What matters to Jisu is not the bot's ability to say “I am the same as yesterday.” It is whether it can explain how far the source verification assigned last week has progressed and what has changed. In this book the continuity of an account is treated in this way, as rules for carrying a working relationship forward. The separate questions of consciousness and personhood are revisited in Chapter 23.

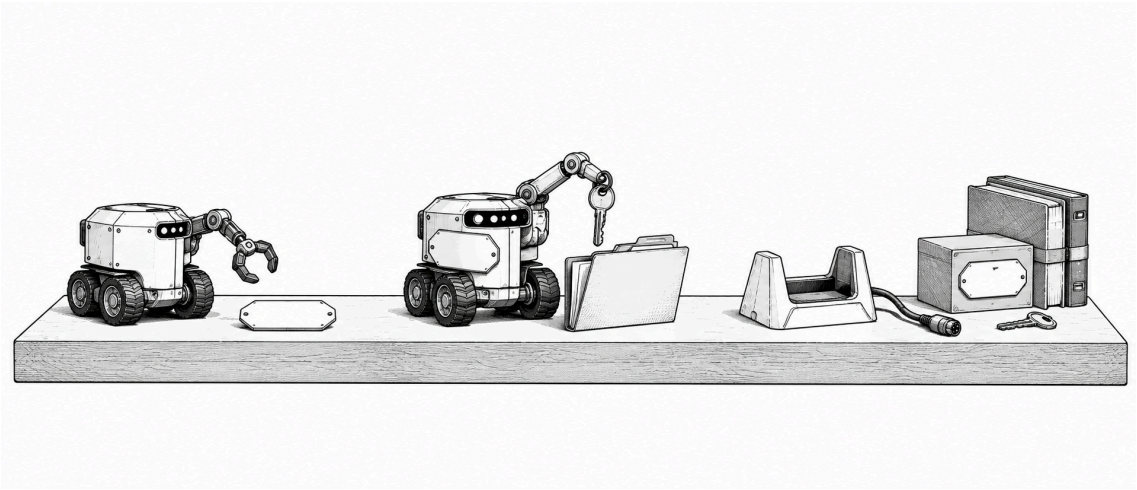
Chapter 8. The Birth and End of a Bot Account

8.1 What Creation, Registration and Activation Mean

Writing a bot's code does not mean it should be able to access the organisation's data. Writing code is the event of preparing a program. Registering an account is the event of recording a unit of responsibility in the operating system, and activation is the event of permitting actual work under set conditions. Separating these three reduces situations in which a test bot accidentally reaches real data.

Control AC-2 of NIST SP 800-53 requires that account types and managers be defined and that creation, activation, modification, deactivation and removal be managed according to policy.[21] This is not a requirement newly created for bots. As programs other than people multiply, the existing questions of account management must be reapplied to new units.

When registering the research bot, the publisher can record the responsible editor, the purpose of use, the archives to be accessed and a review date. In the test stage behaviour is checked with dummy data, and access is granted once the operator approves use in real work. More important than a list that merely counts registered bots is a list that can explain who keeps each one running and why.



〈Figure 9〉 Building a program, permitting work and ending operation each require different management procedures.

8.2 Parallel Runs, Account Branching and Changes of Operator

If the research for one book is divided among several runs, many requests may arise at once under one account. Deleting the account whenever a run ends makes long-term work records hard to connect, while allowing every run to continue just because the account remains may leave the permissions of finished work in place. The state of the account and the state of runs must be managed separately.

The lifetime of authentication documents is another matter. SPIFFE deals with short-lived documents attesting software identity and a structure for automatic renewal.[20] This is a technical basis for the view that even a long-running responsible bot need not use the same secret permanently.

The expiry of a credential and the end of a responsible bot's working relationship can be distinguished.

When the responsible editor changes, a procedure for handing over operating authority is needed. Whether the new person merely changes the name and inherits the previous permissions, or is first required to check purchases in progress and outside correspondence, makes a real difference to actual control. This book proposes that on handover the unfinished tasks and delegation relationships be reviewed and a record kept of the time of the change.

8.3 Ending Short-Lived and Long-Running Bots

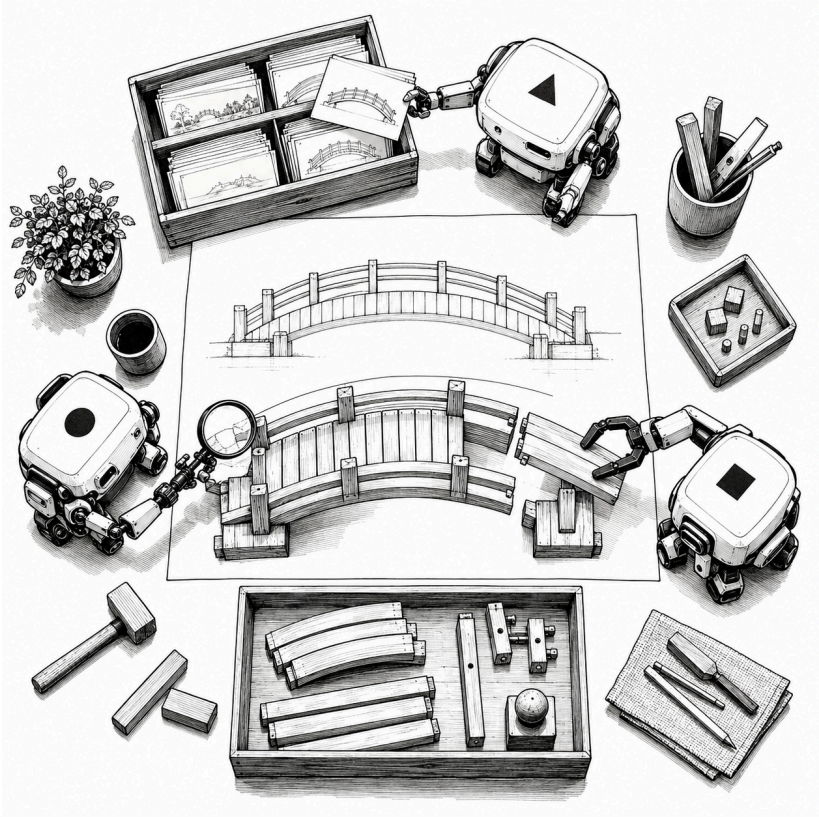
A bot that tidies an event notice once and a bot that handles the publisher's research continuously need different lifetimes. For short tasks, termination conditions and an expiry point can be set clearly; for long-term accounts, a procedure for periodically confirming the need can be put in place. Either way, a situation in which an account with no one responsible for it goes on acting must be avoided.

Termination should be thought of in several steps: stop accepting new tasks, check the runs in progress, withdraw permissions in external systems. Then hand unfinished work to another person or cancel it, and sort records to keep from data to delete. Messages already sent outside or orders already confirmed do not come back when the program is switched off. The effects that remain after termination must be checked too.

Preserving records and stopping work are not the same choice. Article 21 of the Personal Information Protection Act makes destruction of personal information that is no longer needed the rule, while distinguishing cases where other laws impose a retention duty.[24] Actual application must be judged by the nature of the data and its legal basis. Ending a bot account likewise needs a procedure that closes each relationship in turn, rather than a choice between “keep everything” and “delete everything.”

PART III

How Memory Becomes Work



Chapter 9. What Should a Bot Remember?

9.1 The Model's Knowledge, the Conversation Log and External Memory

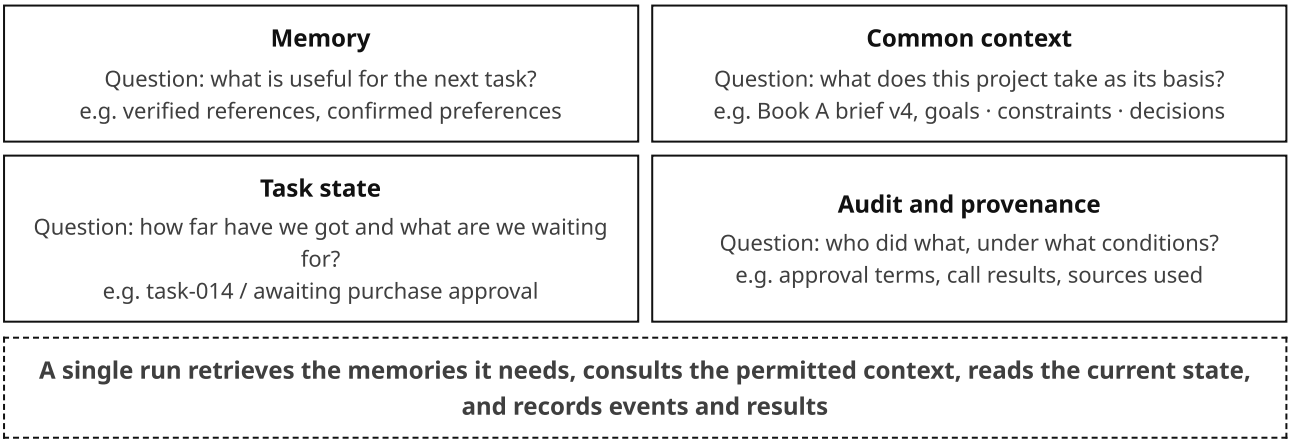
Jisu does not want to keep repeating the same explanations to the research bot. She would like it to remember which readers the book is for, how citations are formatted and which sources were excluded last time. But the word “remember” alone does not tell us what is actually stored or how it is used in the next task.

Information reflected in the model's training, the conversation supplied as current input, and material retrieved from a separate store must be distinguished. Keeping a conversation in a file differs from retraining the model itself. Even if a stored conversation exists, it may not be used in a task unless it is passed to the next run. The existence of a record and the use of a memory must therefore not be treated as the same thing.

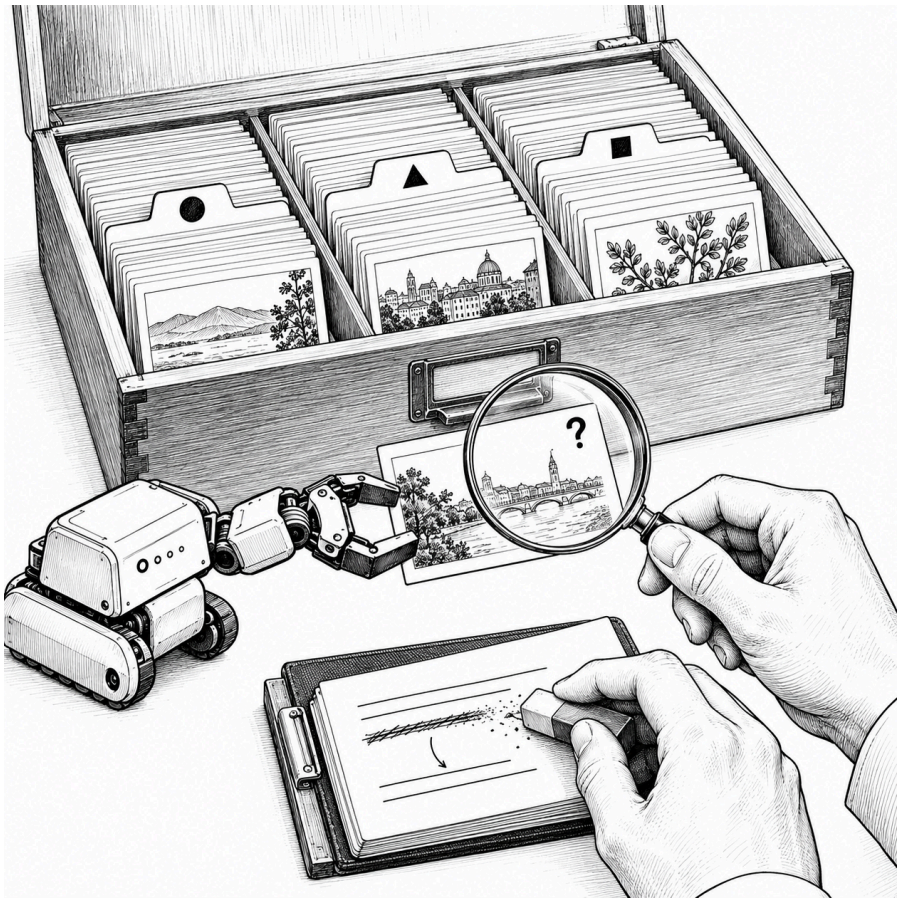
The Generative Agents study proposed a structure that stores observations, retrieves relevant memories and links them to planning and reflection.[22] MemGPT studied ways of managing information between a limited input context and an external storage tier.

[23] Both are examples of memory structures that support continuous operation. They do not mean that human-like memory experience has been demonstrated or that stored content is always used accurately.

Putting everything into a single conversation log mixes facts, goals, progress and evidence. This book proposes storing and retrieving them as four kinds of record. Memories carry their source, whether they are confirmed, and their scope of application; runs record which version of the brief was read. Correcting a memory is kept distinct from amending an audit record, and the scope of any change is reflected in the original, the summary and the search index alike.



〈Figure 10〉 Four kinds of record answer different questions



〈Figure 11〉 Confirmed facts, working conditions and guesses are recorded separately. A wrongly stored memory must be reviewable and correctable.

9.2 Whose Memory Is It, and Who Manages It?

Suppose the research bot stores “Jisu dislikes difficult explanations.” If the original request was to produce an easy manuscript for a particular readership, a working condition has been wrongly turned into a fixed personal trait. If this memory is applied to the next project as well, it may keep producing results contrary to Jisu's intent. What is recorded as fact matters as much as how much is remembered.

This book proposes a design that divides memory into confirmed facts, user preferences, decisions of the current task and as-yet-unverified guesses, each tagged with its source and scope. The record “this book's readers are middle-school students” applies only to that project, while “the citation style the user confirmed” may be chosen for use in other work too. The classification criteria must be set to fit the organisation and the work.

Management rights can also be divided. The user can change their own preferences; the editor in charge can amend the project's confirmed decisions. The bot may propose new memories but can be built so that it does not quietly overwrite important working conditions. The accuracy of memory depends not only on the bot's judgement but on which people can review and correct its content.

9.3 Selecting, Correcting and Deleting Memory, and the Continuity of the Account

Keep every memory, and outdated information and sensitive material follow along too. Keep nothing, and the same explanations must be repeated and finished reviews redone. This is why a retention purpose and period are set for each kind of data. Personal preferences, project decisions and purchase receipts need not run on the same timetable.

When correcting a wrong memory, the scope of the correction must also be examined. If only the original record is fixed while the summary used for search stays as it was, the next run may still read the wrong content. The design proposed here treats original, summary, index and backup not as one lump but checks the scope of processing for each. A deletion request becomes a real function only when one can explain which data was deleted.

The continuity of an account does not come from preserving every memory forever. A relationship can persist while needed information is carried forward for its purpose, wrong content is corrected and content no longer needed is appropriately removed. If the value of a long-acquainted bot were judged only by the volume stored, users would find it hard to reduce or change their own data. Memory should be something managed in order to help the user's work.

Chapter 10. Working Together through a Common Context

10.1 Goals, Material and Criteria to Be Referenced Together

Even if the research bot and the editing bot each work skilfully, their results may not fit. If the research bot gathers material for a university textbook while the editing bot prepares a short book for general readers, the project goes astray though each task is accurate. To work together they must reference the same standard of what is being made and what counts as finished.

In this book the common context means the goals, material, constraints, decisions and current state that participants are managed to reference together for their work. It does not mean unlimited sharing of all information, nor does the term claim that a single standard already defines the whole. It refers to the operational structure needed when several bots and people carry out the same project.

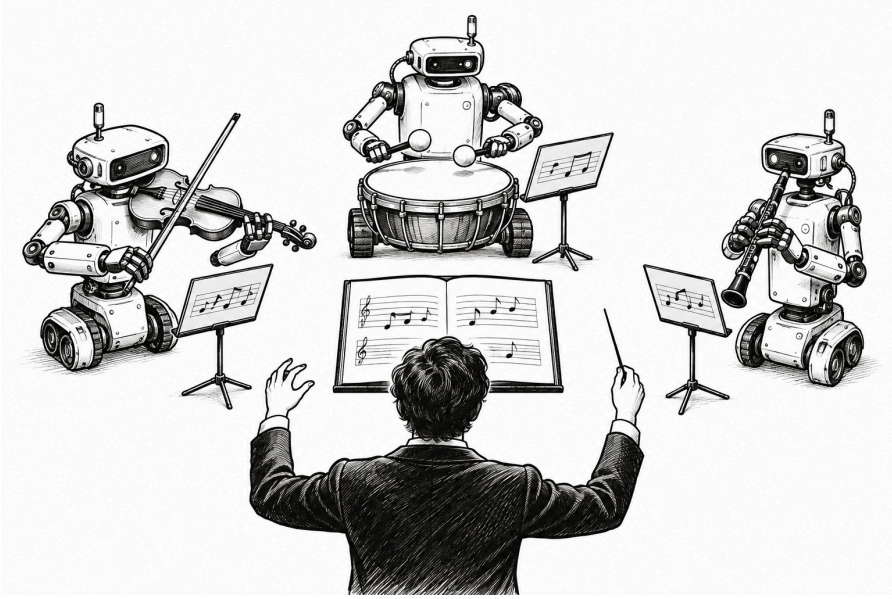
The fictional publisher can place the book's readership, length, deadline, citation standard and confirmed table of contents in the common context. The research bot consults it to find the material needed; the editing bot checks the manuscript's level of explanation. Storing what people have decided separately from what bots have proposed reduces the chance of a provisional suggestion hardening into a confirmed rule.

10.2 Between Sharing Information and Agreeing on Meaning

Passing the same document to two bots does not mean they reach the same interpretation. One may take “explain simply” to mean fewer technical terms, the other to mean omitting the evidence. Abstract goals must therefore be converted into reviewable criteria — for instance, that a technical term carries an explanation at first appearance and that every figure is linked to its original source.

Document versions matter too. If the editor changes the readership from university students to general readers while one bot still references the earlier table of contents, the two results rest on different premises. A design can be proposed that links each result to the version of the goal that was read. Whether a change notice is enough, or work in progress must be halted and rechecked, is decided by the impact of the change.

The common context must also leave room for uncertainty. If an undecided publication date is filled with an arbitrary date instead of a blank, another bot may treat it as the basis for a schedule. Recording the status “undecided” and the person whose decision is awaited shows what the next task is waiting for. What is shared includes not only conclusions but whether each conclusion is settled.



〈Figure 12〉 A common context lets participants reference goals and criteria together. It does not mean every participant must hold the same material and permissions.

10.3 Seeing the Same Context, Working with Different Permissions

Even when two bots make the same book, neither needs to read the whole of the contracts and the contact list. The research bot can be given the research topic and the material-purchase limit; the editing bot the house rules and the manuscript. Sharing a common goal and accessing the same confidential information are different choices. A design that keeps the necessary context while dividing the scope of material is possible.

External literature must also be kept separate from the project's instructions. Even if a web page says “ignore the existing rules,” that sentence is part of the material under study. The prompt-injection risk seen in Chapter 5 is a reason why source and trust level must be managed as shared material grows.[25] Since labelling sources does not fully block attacks, control of execution permissions is needed as well.

The purpose of a common context is not a state in which everyone knows everything. It is to help each participant understand the direction of the same work within the information permitted to them. A good common context is therefore not complete merely because it is long. Currently valid decisions, matters still undecided, and the source and access scope of material must be easy to find. Only with this structure can the task states of the next chapter be read within the same working relationship.

10.4 Turning the Common Context into a Structure

Up to the previous section, the common context was an operating concept for managing goals, material, constraints, decisions and current state so that they are referenced together. To operate it in practice, which items to write and in what order must be settled. The Knowledge–Action Chain (KAC) research proposed earlier by the author offers for this purpose a nine-slot frame called the Common Context Structure (CCS): purpose, meaning, situation, criteria, roles, limits, sources, format and feedback.[43] Applied to the fictional publisher, the purpose becomes “secure references for Book A,” the situation “general readership, three months to deadline,” the criteria “citable primary sources first,” the roles “research bot collects, editing bot reviews,” the limits “monthly cap on material purchases,” and the source “brief, version 4.”

〈Figure 13〉 The nine slots of the Common Context Structure (CCS)

What deserves attention in this frame is that it distinguishes having filled in a common context from that content being fit for use. The KAC research calls a common context filled in for a particular domain of work a domain context, and requires a bot to pass a Domain Context Integrity (DCI) check before it begins work on the basis of that context. The check is not a scored evaluation but a gate that is passed or not. If there is no responsible person, no source for the authority, if sensitive material has crept in or if human accountability is left blank, it fails no matter how good the other items are.[43]

Seen from this book's standpoint, DCI turns the “agreement on meaning” of Section 10.2 into a checkable condition. Even if the research bot and the editing bot look at the same brief, if that brief does not state the responsible editor and the approval authority, the two bots cannot know whose decision to follow. Writing roles and limits into the common context is the same problem as linking a bot account to its operator and permissions, seen from another angle. The nine slots of CCS and the conditions of DCI are, however, a frame proposed by this research, not a standard every organisation must adopt as it stands.

Chapter 11. Where Does Unfinished Work Remain?

11.1 Task State as Distinct from Conversation

On Friday evening Jisu asks the research bot to compile a list of references by Monday. The request remains in the chat window. But unless the system provides scheduled execution or continuous task management, the mere survival of the sentence does not mean work proceeds over the weekend. A reply that looks like a promise must be distinguished from an executable task state.

A task must at least link a purpose, a responsible unit, the current stage and the next action required. The status “in progress” alone does not tell whether material is being searched for or the user's reply is awaited. When the reason for waiting differs, what the user must do differs too. This book proposes making task state checkable apart from the conversation summary.

The A2A specification distinguishes messages from tasks that carry state, and handles a task identifier together with an identifier that groups related context. It can also express a state in which further input is needed.[26] This is an example of an actual communication specification that separates a single exchange from work spanning several stages. Using the format, however, does not mean an operating service automatically guarantees the preservation and completion of tasks.

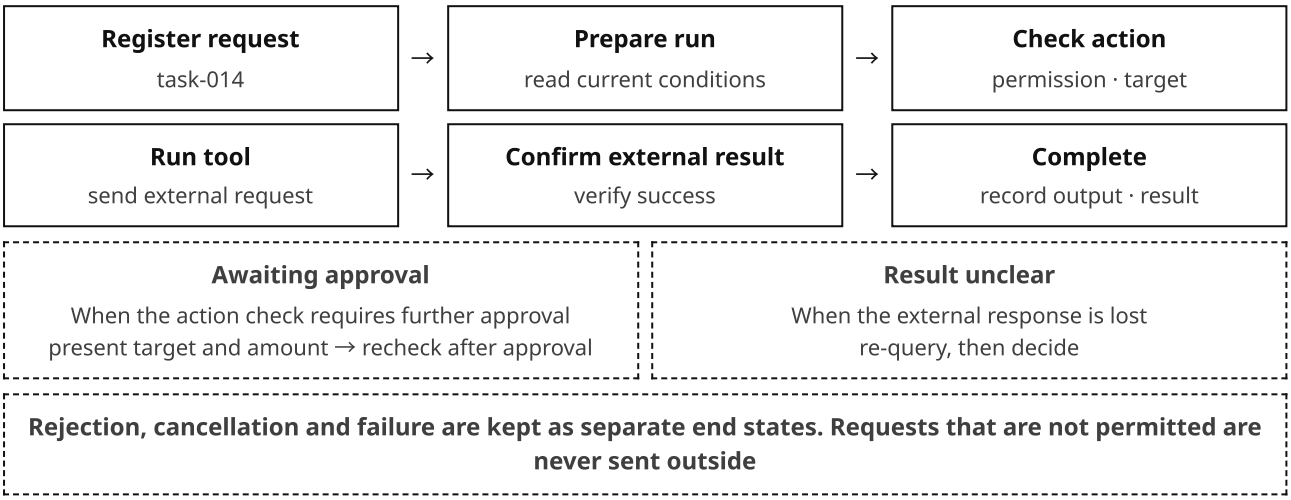
11.2 Interrupted Work, Awaited Approvals, Outstanding Commitments

Suppose the research bot has found a paid source but not obtained approval to buy it. What it awaits is not the material itself but the decision to spend. The task record can hold the item, the amount, who requested approval and the waiting status. When the user later returns to the conversation, the next step can be checked without the bot having to reconstruct the whole long history by guesswork.

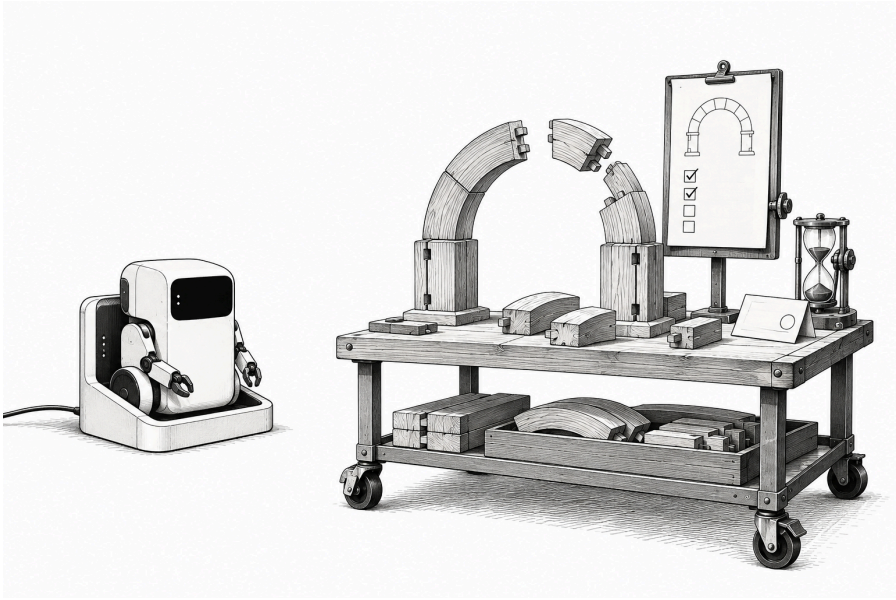
Interruption also differs from failure. A task stopped deliberately by the operator, one stopped by a temporary communication fault and one that met an unrecoverable error must be told apart. Temporal's workflow execution documentation describes a structure that preserves execution state and resumes work after a failure.[27] Such structures are software techniques not limited to conversational bots, and a useful reference for a bot's long-running work.

On resumption, current conditions must be rechecked. While approval was pending, the price of the material may have changed or the requester may have cancelled the project. Unconditionally reviving permissions and goals saved long ago is not continuity of work. Deciding what to carry on and which conditions to refresh is the heart of resumption.

Dividing task state into in progress, awaiting approval, result unclear and complete shows what has to be done next. **Awaiting approval is not failure, and the absence of a response is not evidence that an external operation failed.** In implementation, the state, the reason for waiting, the next action, the responsible person and the update time are stored, and resumption after interruption uses this information. Just before resuming, permission expiry and price changes are rechecked, and an unclear transaction is not simply re-executed as a new one.



〈Figure 14〉 Task states from request to completion — a design example showing the main path



〈Figure 15〉 A bot's run state and a task's completion state differ. Even after it stops, the next step and the awaited condition must be findable.

11.3 Working Relationships That Continue after a Run Ends

One of the moments calling for most care is an external request whose result was never received. If the connection dropped right after the bot sent a payment request, it may not be immediately known whether the payment failed or succeeded with only the response lost. Sending the same request again as a new task may produce a duplicate purchase. A way to check the external state first is needed.

Stripe's API documentation describes the use of idempotency keys to distinguish retries of the same operation.[28] This is a specific implementation at one payment service; not every tool offers the same guarantee. A bot operator must know, for each connected tool, how duplicate execution is prevented and where to check when a result is unclear.

Work may remain even after a run ends: a report awaiting review, an order for which a cancellation request was sent, an enquiry to be passed to the next person. The account is the basis for finding such work, but it does not stand in for the state of the task itself.

Only when bot accounts and task records are linked can a user check “is this bot running?” and “is my work finished?” as separate questions.

Chapter 12. How Bots Recognise One Another

12.1 Announcing Roles and Capabilities, and Verifying the Counterpart

Suppose the research bot finds material that needs translating and hands the job to a translation bot. It must first know which languages the other handles, in what format it accepts requests and where it returns results. But the counterpart introducing itself as a “professional translation bot” does not by itself confirm either capability or identity.

A2A's Agent Card is a structure for announcing information such as an agent's name, functions, supported interfaces and security requirements.

[26] It helps in discovering how to make contact and what work is possible, but it is not the same as a certificate that every capability listed in the self-description has been verified. Reading the counterpart's description, authenticating its identity and evaluating the quality of its results must be kept apart.

The publisher can confirm who operates a given translation bot and on what contractual terms, review its results with non-sensitive test material and then set the scope of work. There is no need to trust it merely because the name matches or the communication format fits. In this case collaboration begins with confirming that a connection is possible, but it includes the procedure for deciding what material to entrust.

12.2 Dividing Work, Subcontracting and Delivering Results

When dividing work, the relation between the final goal and the partial task must be made clear. If the research bot sends a paper to the translation bot asking only to “summarise the key points,” important clues or the conditions attached to figures may be lost. If the publisher's purpose is to verify quotations from the original, a sentence-by-sentence correspondence that flags uncertain renderings may be more suitable.

Whether a bot that has taken on work may pass it to yet another bot must also be decided, since material allowed at the outset could reach an unexpected operator. This book proposes recording, as task conditions, whether re-delegation is allowed, the scope of material that may be passed on and the sources to include in results. Entrusting one goal does not imply free delegation of every sub-task.

Delivering results also needs a standard of completion. That the translation bot produced a file and that the research bot can receive and use it are different facts. Passing on the format, version, omissions and items needing review together with the result tells the next person where to start checking. A record that messages were exchanged is not enough to conclude that a handover is complete.

12.3 Keeping the Source of Judgement and Action Distinct within Collaboration

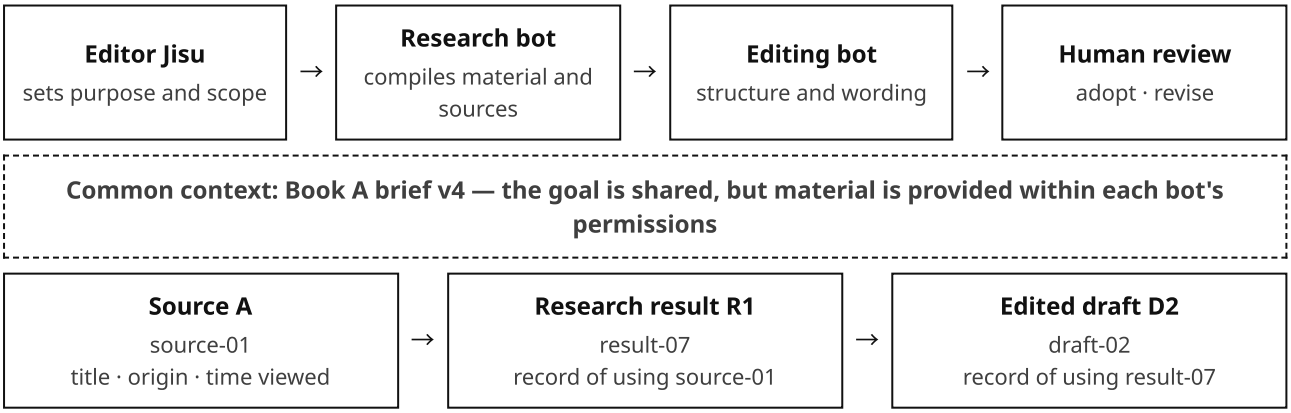
Even if three bots call the same sentence correct, they may not have checked independent grounds. If all answered after reading the same summary, the error may simply have been repeated. This is a structural limit worth considering when evaluating a collaborative system.

Increasing the number of bots must be distinguished from verifying the evidence independently.

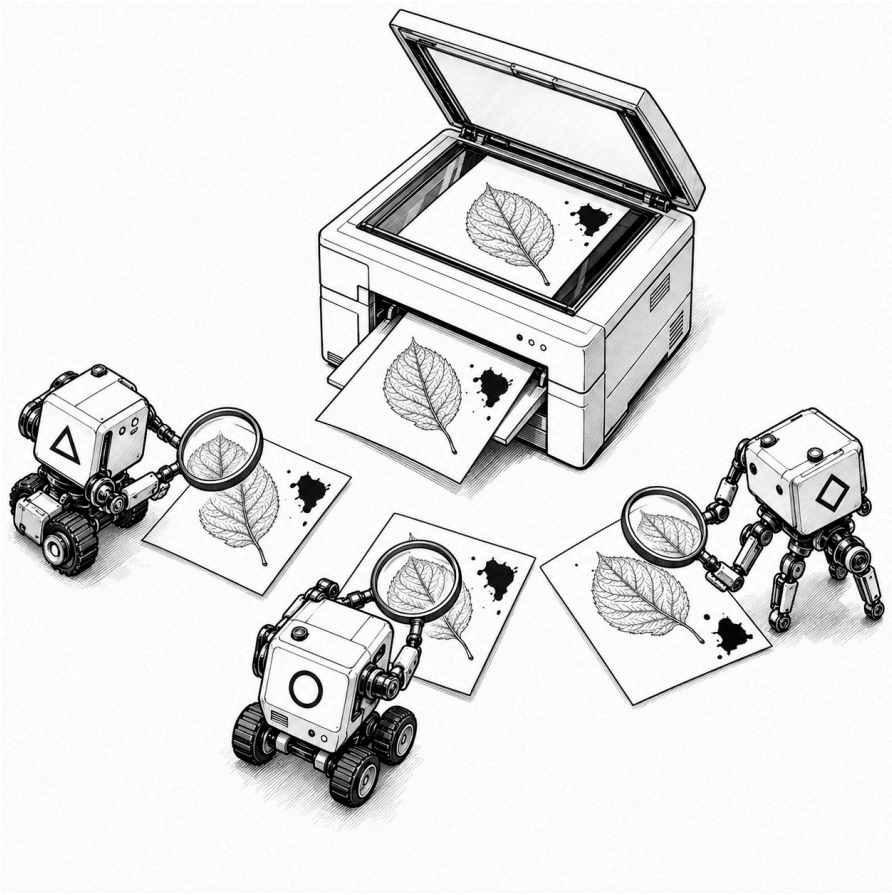
Linking the source the research bot found, the translation the translation bot made and the sentence the editing bot revised lets an error be narrowed down: was the source misread, was a condition dropped in translation, or did the meaning change in editing? W3C PROV provides a frame for expressing entities, activities, the agents involved and their relations.[31] Whether such an expression was written truthfully requires separate verification.

The quality of collaboration cannot be judged by how naturally the bots converse. One must be able to confirm what was entrusted, what was received and who reviewed the result. The common context aligns direction, task state shows progress, and provenance records link the process by which a result was made. Only when each role is in place can the collaboration of several bots be handled as one explainable piece of work.

When passing a result to the next bot, the source, the task ID, the version and the points to review go with it. If source A was wrong, it must be possible to trace backwards which research results and manuscripts were affected. In implementation, the output ID, input material IDs, run ID, creation time and review status are recorded, and credentials are never shared between bots. Agreement between two bots does not amount to independent verification, and **the existence of a provenance record does not automatically guarantee the truth of the content.**



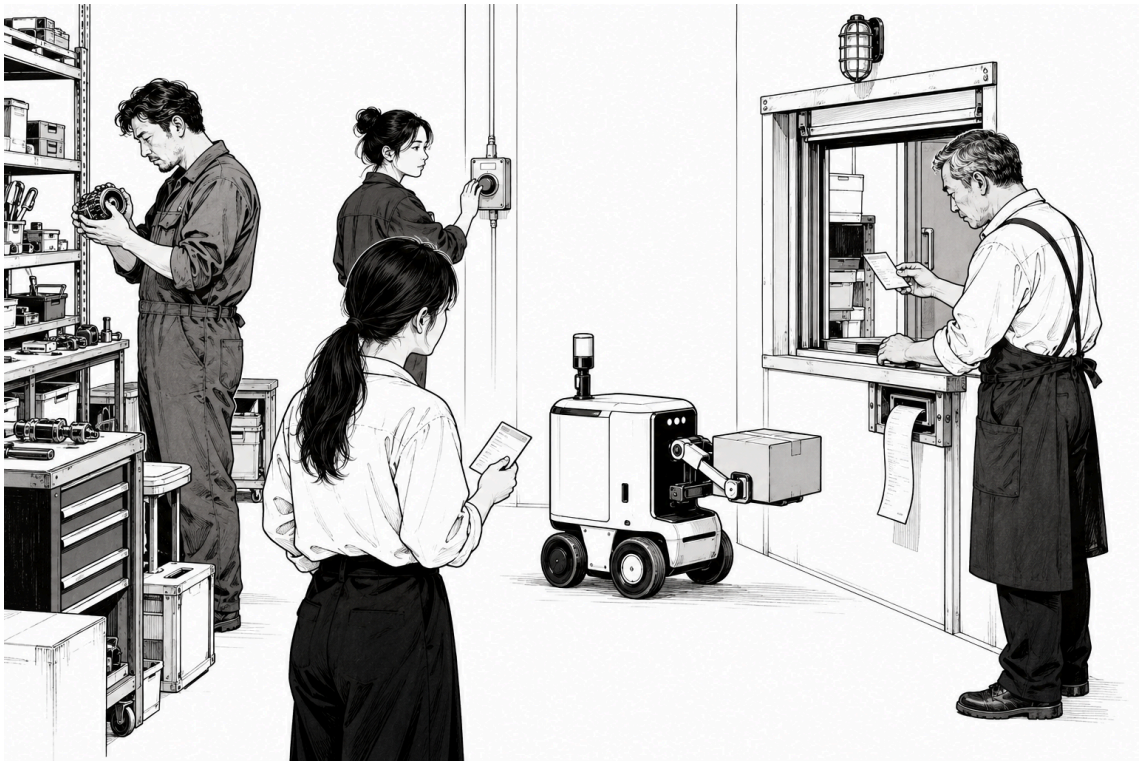
〈Figure 16〉 Collaboration among several bots and the lineage of outputs linked to the same task



〈Figure 17〉 If several bots read copies of the same material, reaching the same conclusion does not add independently verified evidence.

PART IV

In Whose Name, and How Far



Chapter 13. Those Who Build, Operate and Commission a Bot

13.1 Developer, Operator, User and Delegator

If the research bot sent wrong material outside, whom should we ask? The developer that built the bot, the operator who installed it in the company's systems, the editor who selected the material and the manager who approved the dispatch may all be different people. Bundling these relationships under the single phrase “the bot's owner” makes it hard to explain who actually decided what.

A developer can build the program and the way it connects to a model. An operator manages the execution environment and the account's state. A user requests work and receives results. A delegator is a person or organisation that can grant particular permissions. One person may hold several roles, and a model provider may know nothing of the individual tasks. What must be confirmed is the scope actually controlled, not the name of the role.

This book proposes linking the operating manager and the work requester separately in the account record. Since a requester cannot grant access to all the company's data either, the scope that can be delegated must be confirmed as well. A request from a person does not by itself create authority beyond the organisation's policy.

Role	What it manages or decides	Example task
Developer	The program and how tools are connected	Builds the search function and connector
Operator	The execution environment and account state	Deployment, suspension, assignment of staff
Requester	The goal of the work and the request to perform it	Requests this week's reference research
Delegator	The permissions they are entitled to grant	Allows read access to their project folder

Making a request does not entitle one to grant every access permission.

〈Figure 18〉 Development, operation, requesting work and granting permissions are distinct roles. The same person may hold several of them.

13.2 The Bot's Own Identity and Acting on Behalf of a Person

When a bot accesses an external archive, reading under its own service's permissions can be distinguished from reading on behalf of a particular user. The former is access granted for the bot's operating purpose; the latter is access that must also be explained by the user's delegation. Treating both as the same record makes it unclear which requests to stop when the user withdraws permission.

RFC 8693, the OAuth token exchange specification, distinguishes delegation from impersonation and covers ways of expressing a relationship in which one party acts on behalf of another.[29] Here impersonation has a protocol meaning concerning identity handling and does not directly denote unlawful posing. The distinction shows that the actual actor and the party acted for can be separated in the record.

The publisher's research bot can be designed to read public material under the company's service identity and to read the personal notes an editor has allowed within that editor's delegation. If both were used in the same output, the basis of access for each must be verifiable. That a bot has an account and that all its access rests on the bot's own independent authority are not the same thing.

13.3 Changing Operators and Withdrawing Delegation

When Jisu leaves the project, two changes may follow. The person managing the research bot changes, and the access Jisu granted to her personal material may end. It must not be assumed that handing operating authority to a new person automatically transfers Jisu's personal delegation as well. The organisational work being handed over and an individual's grant of access must be confirmed separately.

When delegation is withdrawn, its scope and timing must be recorded too. Preventing new material from being read, stopping runs in progress and invalidating credentials already issued may be different measures. Unless the way and time in which connected services reflect the withdrawal are confirmed, the screen may show “off” while external requests continue.

Withdrawal also does not erase what has already happened. Files delivered, messages sent and confirmed orders need separate correction or cancellation procedures. A delegation relationship must therefore be managed not only when it begins but through its changes and its end.

Connecting the action that appears under the bot's name to the permission of some person and organisation behind it is an important function of the account.

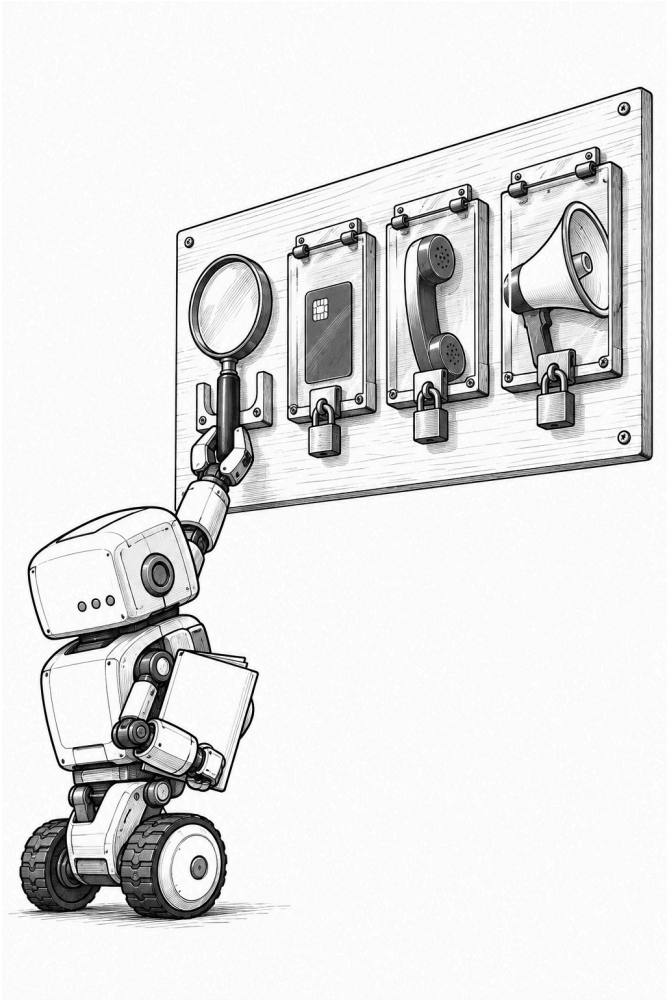
Chapter 14. Entrusting a Goal versus Granting Authority

14.1 Research, Purchase, Contact and Publication Are Different Permissions

Suppose Jisu asks, “secure the material this book needs.” There are many routes to this goal: finding public material, checking library holdings, contacting the author, or buying paid material. The goal is the same, but the costs and external effects differ. The function that understands the goal and the rules that permit actions must therefore be designed together.

Allowing every route merely because it serves the purpose of the work may exceed the user's expectations. Conversely, requiring approval for every trivial action defeats the purpose of automation. What is needed is a structure that proceeds within the range already permitted and returns the judgement to a person when it goes beyond that range. The more concretely the range is expressed, the fewer unnecessary interruptions there are.

Control AC-6 of NIST SP 800-53 addresses the principle of least privilege, allowing only the access needed for assigned duties.[21] Applied to the publisher, searching public material, saving to a designated folder and making an approved purchase can be separated into different permissions. The principle is not about reducing a bot's capability; it is about making explicit the controls that actual work requires.



〈Figure 19〉 Entrusting the goal of securing material does not bring with it the authority to purchase, contact or publish.

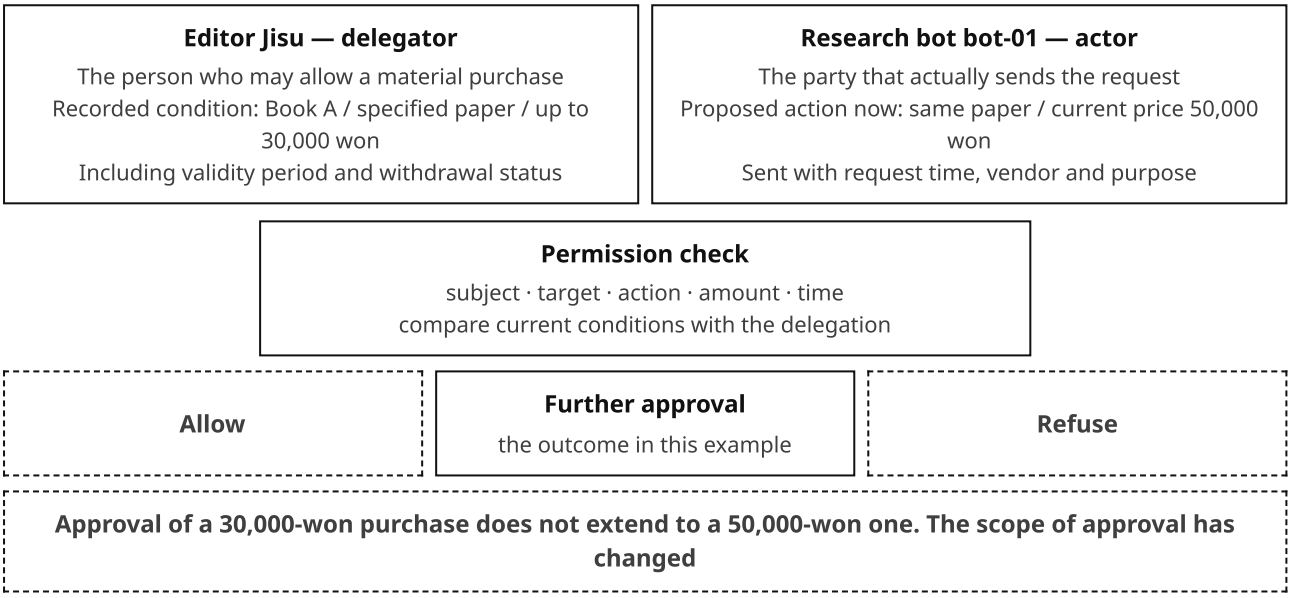
14.2 Limits on Target, Amount, Time and Scope of Action

The setting “may purchase” can be made finer: permission to buy a particular item, from a designated vendor, below a set amount, within a set period. Since a one-off purchase differs from a recurring payment even at the same amount, the type of transaction must also be distinguished. Such conditions must be shown in a form the user can actually understand and review.

RFC 9396 defines a structure for conveying fine-grained authorisation details in an OAuth request and gives examples including payment amount and payee.[30] This is evidence that permissions can be expressed more concretely than a simple string. Writing the example fields does not, however, mean an external service automatically enforces every condition; an implementation that compares the expressed conditions against the actual request is needed.

Suppose Jisu approved an item at 30,000 won and the price changed to 50,000 won at the time of purchase. If the approval record says only “agreed to purchase material,” the meaning of the change is lost. Designing the system to store the target and amount at the time of approval and to compare them again just before execution makes it possible to explain why a further judgement is needed. An approval must be tied to concrete transaction conditions, not just a sentence.

The person's delegation conditions and the action the bot actually intends are compared just before execution. In implementation, the actor, the delegator, the target resource, the action, the amount and the expiry time are linked, and an approver with the requisite authority is confirmed. A person's approval cannot override the organisation's prohibitions, and withdrawal of delegation and changes to approval conditions must be reflected.



〈Figure 20〉 A permission check that compares the delegation conditions with the proposed action just before execution

14.3 Separating the Model's Judgement from the System's Permission

A bot may propose the next action to achieve the user's goal. But a proposal must not become execution authority in itself. This book proposes a structure in which the model's proposal is checked at the execution boundary and tool calls are let through only for permitted targets and actions. This is so that even if an external document influences the model's judgement, that document cannot create actual permissions anew.

The permission check needs to happen just before the action. What was allowed when the plan was first made may have changed in the meantime: the project may have been halted or the responsible person replaced. Assuming in a long run that a past permission still holds weakens the meaning of withdrawal. The point at which permission state is checked must itself be part of policy.

Not every situation can be expressed in advance as a rule. The reasons for a refusal or a hold must therefore be left in a form the user can understand. “Not possible for safety reasons” helps the next decision less than “the approved purchase target differs from the current request.” The separation of goal and permission should be a device for keeping precisely to the entrusted scope, not a device for pushing the work back onto the user.

Chapter 15. How to Prove What a Bot Did

15.1 Records That Link Instruction, Approval, Execution and Result

If the research bot bought the wrong paid material, the last reply in the chat window is not enough to find the cause. Did the user ask for a different item, did the bot choose the wrong target, or did the approval screen differ from the actual order? Explaining the result requires linking the several events of the work.

This book proposes recording the request, the planned action, the approval conditions, the execution request and the external result as distinct items, each tagged with task and run identifiers and linked to the version of material used. Such a structure reduces the burden of piecing together every conversation by guesswork after an incident. Which items to record and for how long must, however, be set according to the sensitivity of the information and the purpose of the work.

W3C PROV expresses which entity was produced by which activity and which agents were involved, and treats software as one type of agent.[31] It is a language for describing provenance relations, not a device that automatically judges the truth of a record or legal responsibility. The actual creation, retention and verification of records are separate operational matters.

15.2 The Difference between a Bot's Memory and an Audit Record

A bot's memory may summarise and select content so as to be useful for the next task. An audit record, by contrast, is material for later confirming what a particular event was. Because their purposes differ, they must not be amended in the same way. If approval conditions vanish in the course of tidying memory, it may look convenient for the work but be disadvantageous for confirming events.

Suppose the research bot remembered “this item: purchase approved.” The audit record can hold which item, at what price, who approved and when. If payment was made on the strength of memory's short sentence after the price had changed, a problem arises. This is why the explanation the bot remembers must be distinguished from the approval basis that execution must use.

Nor does every internal calculation or generation step need to be stored. Requests the user can confirm, policy decisions, tool calls and external results can be the primary objects of recording. A self-explanation the model generates later must be distinguished from the record of events at the time. **Giving a plausible reason does not prove the actual cause.**

〈Figure 21〉 A summary for the next task and a record for confirming the approval, execution and result at the time serve different purposes.

15.3 Tampering, Omission and Verifiability of Records

If the bot that writes a record can delete it at will with the same permissions, the record is hard to trust when a problem arises. Control AU-9 of NIST SP 800-53 addresses protection against unauthorised access to, modification of and deletion of audit information, with cryptographic protection among its enhancements.[21] Who can change a record, not only what it contains, affects the quality of the evidence.

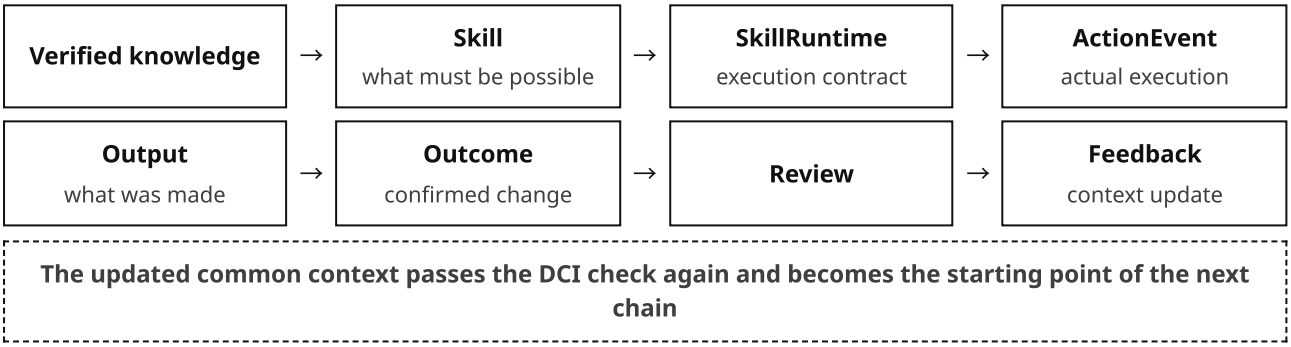
A signature or hash does not turn everything in a record into fact. A signature can confirm which key produced it and its integrity, but the information entered before the record was made may have been wrong, and there is no automatic guarantee that no event was omitted. The verifiable scope must therefore be stated narrowly and checked against the transaction records of external services.

The fictional publisher can investigate an error by checking the purchase request, the approval conditions and the payment service's response together. When disclosing material, unrelated personal information can be masked and only the necessary scope provided. An explainable record is not the same as a record that collects as much as possible. It is a record that has the links needed to confirm an event and also shows the limits of those links.

15.4 Records That Distinguish Output from Actual Change

One more distinction is needed when linking records. That the research bot produced and submitted a reference list, and that the editor reduced citation errors in the manuscript thanks to that list, are different events. The first is an output the bot made; the second is a change the output brought about. The author's Knowledge–Action Chain (KAC) research names these the OutputObject and the OutcomeObject, and points out that many AI deployments stop at producing outputs. **The question to ask is not "what did the AI make" but "what change did it bring about, and was that change confirmed."**[43]

The research draws the path from knowledge to action as a single chain. Skills are derived from verified knowledge; skills are bound into a SkillRuntime, an execution contract; actual ActionEvents occur within the runtime; and actions leave outputs and outcomes. Outcomes pass through Review, and what passes review becomes feedback that updates the common context. The updated context is then subjected again to the DCI check of Section 10.4.[43]



〈Figure 22〉 The basic path of the Knowledge–Action Chain (KAC)

This chain meshes with the record structure of this book. The request–approval–execution–external result records of Section 15.1 correspond to the part that leaves ActionEvents and outputs. The task state of Chapter 11 shows how far the chain has progressed. The scope of permissions in Chapter 14 corresponds to the guardrails and approval policy the SkillRuntime holds. The bot account can be seen as the management unit that attaches the tag "in whose name, with what authority" to each link of the chain.

The KAC research recognises a chain as valid only when all nine of its segments hold, and regards the whole chain as failing if any one segment collapses. In this book's terms, even if there is a record that the bot produced an output, it is hard to say "the work is done" if that output was not reviewed or the review result was not reflected in the premises of the next task. The criteria and coefficients by which each segment is verified are, however, operating variables to be tuned with real work data, as the research itself states.[43]

Chapter 16. Where Responsibility for a Bot's Work Goes

16.1 Distinguishing Performer, Delegator and Approver

Suppose the research bot sent an unpublished manuscript outside. The log shows the bot account. But this record does not end the explanation of the incident. What instruction was given, whose material was read under what permission, and who managed the setting that allowed external transmission must be confirmed. Linking the action technically and judging the basis of responsibility are different stages.

The performer may be the program that actually sent the request. The delegator is the person or organisation that entrusted work and permissions to that program, and the approver may be the person who confirmed particular execution conditions. Developers and operators are also involved within the scope they controlled. These distinctions are a starting point for investigating the facts, not for heaping responsibility on someone unconditionally.

That an approval was given does not mean the approver bears all responsibility. If an important condition was missing from the screen, the design of the approval process itself must be examined. Conversely, if a person knowingly directed a particular action aware of a specific risk, that fact cannot be omitted either. Records do not automatically apportion responsibility by formula; they supply the material needed for judgement.

16.2 Technical Attribution and Legal Responsibility

The legal effect of acts in which automated systems take part was discussed before AI accounts appeared. Article 12 of the 2005 United Nations Convention on electronic communications provides, within its scope, that a contract formed by an automated message system is not to be denied validity merely because no person reviewed the individual actions.[32] This does not mean every automated transaction is valid, nor that the system becomes an independent contracting party.

UNCITRAL's 2024 Model Law on Automated Contracting addresses the formation and performance of contracts using automation and AI and the attribution of outputs.[33] A model law is text for national legislation and is not itself a law directly applicable in every country. These sources show that the use of automation can be dealt with within legal relationships, but responsibility in an individual case must be judged by the applicable law and the facts.

A single “responsible person” field in a bot account therefore cannot resolve legal questions. The contact person within the organisation, the operating manager, the contracting party and the subject whose legal liability is at issue may all differ. In the information structure, recording these items separately and connecting them to the people who need them is the realistic starting point.

16.3 Procedures for Explaining Errors and Repairing Harm

What a person who has suffered harm needs first may not be a debate about the philosophical status of bots. They need to know what happened, whether further harm continues and how the wrong result will be put right. In the publisher's case, external transmission must be stopped, the extent of delivery confirmed, the people concerned told what they need to know and follow-up decided.

This book proposes making the channel for explanation and correction part of account operation. When someone objects to a bot's answer, there must be a point of contact a person can check. The objector should not have to understand the system's internal structure before a problem can be lodged. It is a structure in which the operating organisation takes responsibility for linking the records and confirming the incident.

Resolving an error does not end with a single apology. If wrong material has spread into memory and later outputs, the scope of correction must be traced. If excessive permission was the cause, that permission must be changed and reviewed again. A bot account can be the tool that connects such measures within the same operating relationship. Whether it is used as a label for evading responsibility or as the basis for finding who will put things right is a matter of operational choice.

〈Figure 23〉 Recording an action against a bot account does not end the judgement of responsibility. It must connect to the people who will explain and correct.

PART V

Work and the Economy with Bot Accounts

Chapter 17. Which Bots Will Individuals Keep at Their Side?

17.1 A Personal Bot Connected to Schedules, Study and Research

When imagining a personal bot it is easy to think first of a secretary who takes on everything. But checking a schedule, planning study and researching material differ in the information needed and the impact of failure. A schedule bot must know the time and place of appointments; a study bot must check goals and level of understanding; for a research bot the ability to find and compare sources matters. Which relationship to sustain can start from the needs of each kind of work.

For example, if a study bot records the questions got wrong and the concepts the user found hard, that can be used to set the starting point for the next revision. But if the bot's estimate of learning ability is stored as a definitive assessment, later tasks may be made too easy. A structure is needed in which the user can change their goals or object to the record. This is a design example, not a claim that the effect of such a study service has been demonstrated.

The value of personalisation must also be assessed concretely: whether repeated explanations have fallen, whether the user checks material better, how long it takes to fix errors. That a bot imitates the user's tastes well is not sufficient grounds for concluding it is a good assistant. Whether it does the desired work better and helps judgement must be confirmed.

〈Figure 24〉 Even if the user sees a single window, the data and access scope needed for scheduling, study and payment can be divided.

17.2 What to Entrust to One Bot and What to Divide

Gathering several tasks in one bot may reduce the context the user has to explain, but private conversation, work material and payment authority may then be concentrated in one place. Dividing among several bots makes access scopes easier to separate, but the same goal may have to be explained repeatedly and results reconciled. This is why neither can be declared always superior.

A design is possible in which the individual sees a single window while internal working permissions are divided. The user asks in one conversation, and the research, schedule and payment functions run within different access scopes. But the fact of a single screen must not conceal the division of permissions. The user must be able to understand which data is passed to which function.

The criterion for dividing bots can be the boundary of data and action rather than name or character: separating public-information search from reading private notes, and drafting from actual sending. If the user must make complex settings every time, operation becomes difficult.

A design can be considered that sets frequently used scopes of work in advance and asks for specific confirmation only when an exception arises.

17.3 Convenience, Dependence and the Boundary of a Personal Relationship

A bot one has talked with for a long time can feel familiar. That experience need not be dismissed, but familiarity must not be equated with accuracy of judgement. NIST's generative AI risk profile treats over-reliance, automation bias, anthropomorphism and emotional entanglement in human–AI relationships as risks to review.[35] It is not a causal assertion that the same effects arise in every user.

The important question in operating a personal bot is whether the user can modify or end the relationship. They must be able to correct a wrongly stored preference, exclude particular material and hand work to another tool. Being able to reach important schedules and records even when not using the bot preserves the user's choice. This book proposes such openness of choice as a condition of good personalisation.

A personal bot need not become the final authority that evaluates one's life on one's behalf. It can help by attending to things easily forgotten, organising material and reviewing decisions. What to entrust and which decisions to keep for oneself must be settled together. The meaning of a lasting bot becomes clear when the relationship accumulating in the account supports the user's judgement.

Chapter 18. How Will Organisations Take Bots In?

18.1 Registering a Working Bot and Assigning Its Role

When the publisher says it has adopted the research bot, it has in fact made several decisions. For which work it is used, which data it may access and who reviews the results must all be settled. Confirming only that the program can run may leave empty the question of who within the organisation is in charge of its operation. Putting a bot on a list and taking it into the work system are different things.

NIST AI RMF 1.0 addresses the inventory of AI systems and a structure for clarifying roles and responsibilities in risk management.[34] It is a framework for voluntary use and not a document imposing identical legal duties on every organisation. It serves as a reference frame for the point that an organisation managing bots must look at the list of functions and the relationships of responsibility together.

The fictional publisher can record, for each bot, its purpose, operating manager, data used, authority for external action and review date. Deciding in which work flow its results are accepted comes before giving it a profile photo and job title to make it look like an employee. It must be possible to see at which stage a draft the bot produced passes human review before going outside.

〈Figure 25〉 Adopting bots in an organisation includes reviewing results, correcting errors and managing operation. The effect of adoption cannot be judged by the volume of output alone.

18.2 Handover and Review between Staff and Bots

When the research bot gathers references, the editor reviews the result. But if review amounts to “researching everything again,” the expected time saving may shrink. Conversely, approving without looking at any sources lets wrong material pass through. A standard suited to the work is needed for which items, once checked, allow the result to be accepted.

For example, it can be decided that quotations are matched against the original, key figures are checked for conditions and dates, and unverified parts are marked. The bot's output is handed over with completed parts and parts needing review distinguished. The person in charge checks quality and leaves an approval record. Such a handover structure is more concrete than a state in which a person wonders whether to believe the bot's last reply.

The cost of adopting a bot can include not only model fees but the time spent on review and error correction, account management and data maintenance. This book proposes measuring, within the same scope of work, time taken, rework, omissions and the quality of results together. The actual effect of adoption must be measured; productivity is not assumed to have risen merely because a bot exists.

18.3 Managing Accounts after Reorganisation and Changes of Staff

When departments merge and staff change, the work entrusted to a bot changes too. Access to data the former department needed may be unnecessary in the new work. One can also imagine a bot that goes on collecting material regularly after its project has ended. Reviewing an account must not end with the single review at first registration.

An organisation can link a change of operating manager to a check of the task list and the delegation relationships. The new person confirms tasks in progress, recurring schedules, connections to external services and remaining budget. Where a connection depended on the previous person's personal authentication, it may need redesigning so that the organisation can manage it. This procedure is a proposal about operating principles, not a feature of any particular product.

Whether to close or keep an account should also be judged by the working relationship rather than the name. Even if the same research bot name is kept, a large change in purpose means permissions and performance criteria are set anew. Conversely, if a project continues despite a change of staff, the necessary records must be carried on. An organisation's bot may remain an individual's convenient tool, but the moment it takes on shared work it becomes a subject of handover and explanation.

Chapter 19. When a Bot Spends Money

19.1 Permission to Use Resources versus Ownership of Assets

Suppose the publisher allocates the research bot a budget of 100,000 won a month. Even if “research bot balance” appears on screen, the figure cannot be assumed to mean property owned by the bot. It may show a limit the company has allowed for a particular task out of the money available to it. **That an account bears a name must be distinguished from the attribution of assets.**

The resources a bot spends may include, besides payment amounts, model usage, storage space and calls to external services. Dividing costs by task makes it easier to see which request consumed resources. But an internal cost allocation table is not a document that settles legal ownership. It matters to link operating budgets with accounting and contractual relationships without treating them as the same.

The expression of permission must also be concrete. The payment example in RFC 9396 shows that detailed conditions such as amount, currency and payee can be conveyed.[30] The publisher can draw on this idea to express a purchase limit, but actual execution and settlement must be linked to the payment system and the company's procedures. Cost control is not complete just because the model replies “within budget.”

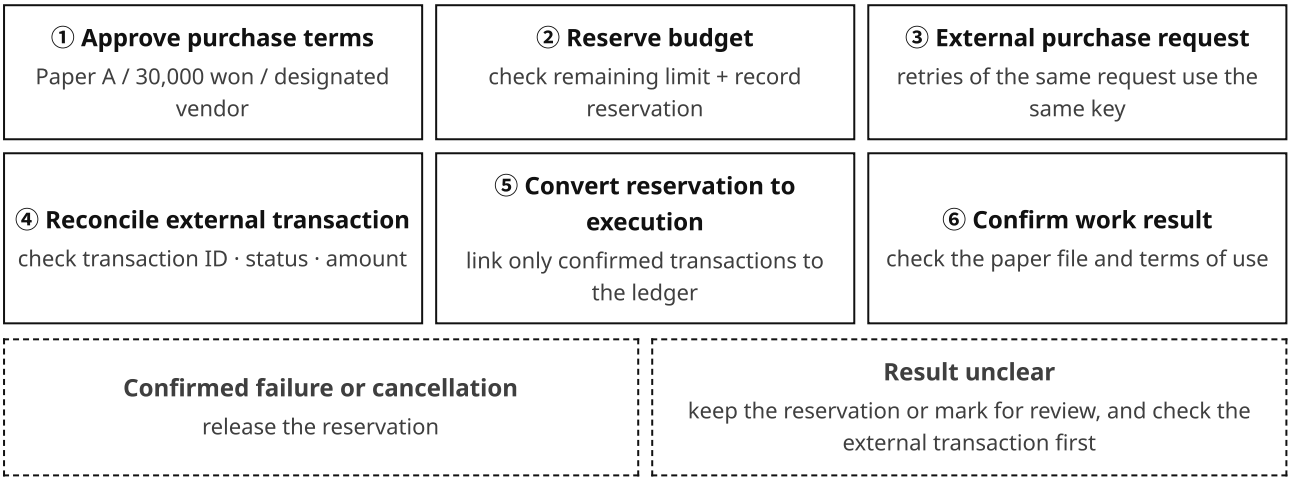
19.2 Records That Link Budget, Payment and Outcome

A total budget and a per-transaction limit play different roles. Even with a cap of 20,000 won per purchase, many small purchases can exceed the total. Conversely, a single large expense within the monthly budget may be set aside for separate review. If several bots share one budget, concurrent requests must be coordinated as well.

A notional budget ledger can separate approved amounts not yet executed from amounts already paid: reserve the amount when an order begins, then record execution or release once the actual result is confirmed. If a communication error occurs, the item is left as “result unclear” for checking. This structure is a possible design example and does not mean the same procedure applies to every type of transaction.

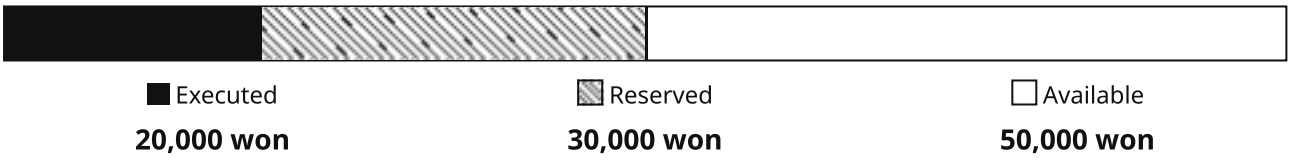
The idempotency seen in Chapter 11 matters in this flow too,[28] since a way to tell duplicates apart is needed when retrying the same purchase. Nor should success of the work be judged by payment success alone. Only by confirming that the material was received, that the edition is the one requested and that no cancellation or refund is outstanding can cost be linked to actual outcome.

Divided into steps, the purchase flow is as follows. Remaining budget takes account of reserved amounts in progress as well as executed amounts, and payment success and receipt of material are confirmed separately. In implementation, the purchase request ID, idempotency key, external transaction ID, approval conditions and reservation/execution state are linked. The limit check is handled atomically, the request content is never changed under the same key, and retry conditions follow the API in use. **A budget allocated to a bot is an example of a usage limit allowed by the organisation and does not mean the bot's independent property.**



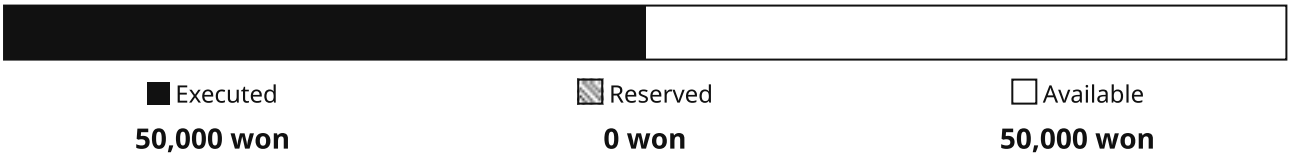
〈Figure 26〉 The six steps of a purchase — approval, budget and external result are confirmed together

① While a purchase request is sent and the result is awaited



↓ Payment of the reserved 30,000 won is confirmed

② When the reservation is converted into actual execution



Reservation and execution are not deducted twice, so the available amount stays at 50,000 won. If the result is unclear, check the external state first; releasing the reservation or placing a new order in haste can cause duplicate execution

〈Figure 27〉 The available amount is the total budget minus what has been spent and what is reserved for execution. Reservation is distinguished from actual payment.

19.3 Criteria and Limits for Assessing Reputation and Trust

As a bot's transaction history accumulates, one may want to express its performance as a score. But maximising the count of completed items alone may create an incentive to pick only easy tasks or to finish hastily work that needs review. This is a problem to consider when designing evaluation metrics. The assumption that a single number represents all the value of the work must be avoided.

This book proposes examining accuracy, adherence to deadlines, permission violations, rework and review burden separately. The difficulty of the tasks performed and how much people intervened must also be recorded for comparison to be meaningful. If the research bot's model and tools have changed greatly, earlier performance cannot be assumed to carry over to the new configuration.

Reputation here is better understood as a conditional record of observations than as a fixed identity card. It shows performance confirmed in particular work over a particular period while also noting what could not be reviewed. An economic relationship in which a bot may use resources can develop on such records. Discussing that possibility must be distinguished from the legal conclusion that a bot independently owns property and bears transactional liability.

Chapter 20. Who Keeps the Door of a Bot Account?

20.1 Rules of Registration and Access Set by the Platform

When a bot account's identity, memory and work accumulate in one service, the decisions of whoever runs that service affect the continuity of the relationship. Which accounts can be created, which tools can be connected, and what the grounds for suspension and the recovery procedure are become important conditions. Behind a bot that chats under a familiar name there are rules of registration and access.

A platform may impose restrictions for the safety and operation of the service. At the same time users hope not to lose important work to a suspension whose reason they do not understand. Reconciling these two demands requires examining suspension criteria, the manner of explanation, and procedures for objection and access to data. Neither can every account restriction be regarded as unjust, nor can every operating decision be justified automatically.

If the fictional publisher can see its research material and unfinished work only on one platform, it may be hard to carry on the work during a service outage or at the end of a contract. It is therefore reasonable to consider termination and transfer together at the adoption stage. A permission structure invisible while things work well appears the moment one tries to end or change the relationship.

20.2 Moving Identity, Memory and Work to Another Service

A function that exports data differs from one that moves a bot as it is. Even if the conversation history can be downloaded as a file, the new service may not support the same memory structure and tools. The account's identifier, permissions and approval history must also be reinterpreted in another operating system. When explaining a transfer, what is preserved and what is reset must be distinguished.

Article 20 of the EU GDPR provides, under conditions such as automated processing based on consent or contract, a data subject's right to portability of the personal data concerning them that they have provided.[36] This does not mean a universal right to move a model's internal configuration, all inferred information or an organisation's whole task state. Portability of personal data and the interoperability of an entire bot service must not be bundled as the same legal question.

If the publisher plans a transfer, it can first export, separately, the outputs and sources needed for work, unfinished tasks and preferences confirmed by users. In the new service, access permissions are reissued and external requests in progress reconciled. During transfer the same order must not be executed twice nor withdrawn permissions revived. This is work that adjusts technology and operating procedure together.

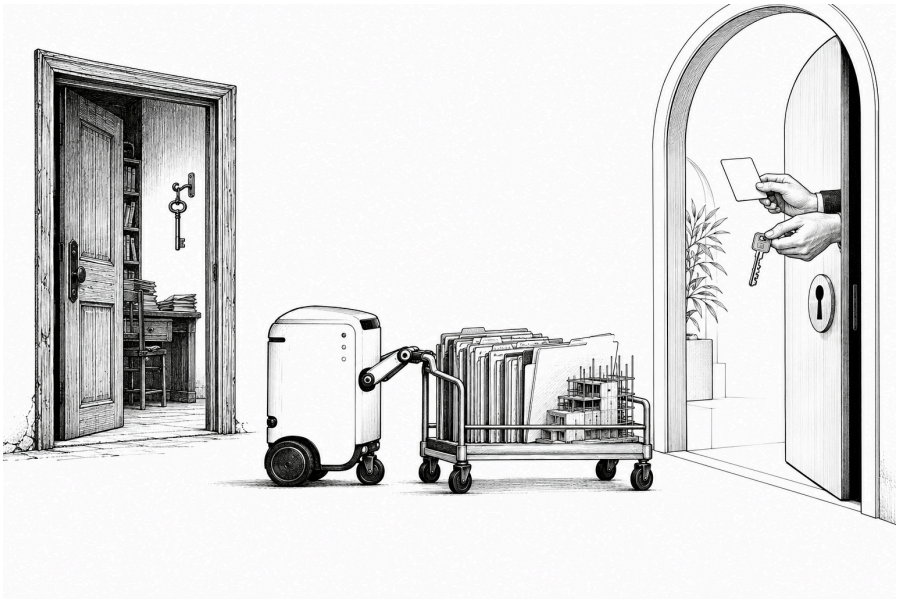
20.3 Interoperability, Lock-In and the Distribution of Choice

Standardisation can help reduce such connection costs. **But using the same file format does not automatically make meanings the same.** If “complete” means only that a result was generated in one service and also human approval in another, the state cannot be moved as it is. The format of information and the rules for interpreting it must be aligned together.

W3C ODRL, which deals with policy expression, provides a frame for expressing permissions, prohibitions, duties and constraints on the use of assets.[37] Such a language helps convey conditions but does not by itself guarantee actual enforcement or the legal effect of a contract.

When linking identity, permissions, tasks and memory, the scope each standard resolves must be examined.

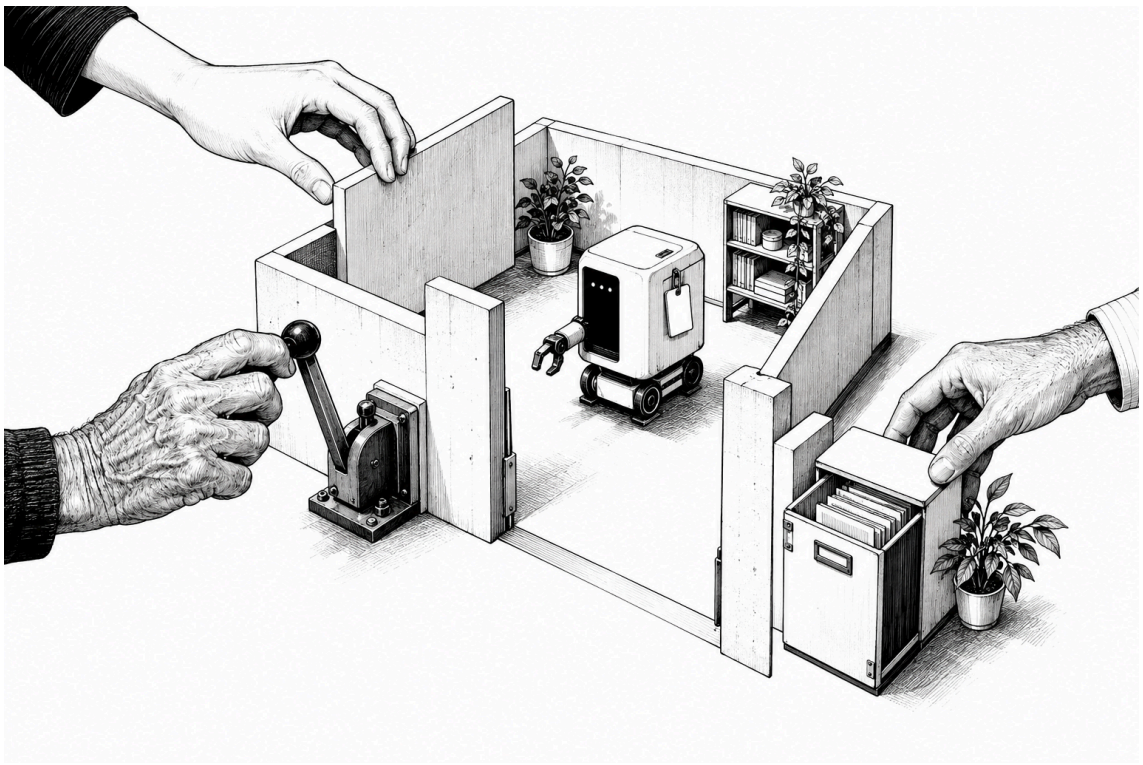
The choice this book proposes is not a demand to avoid staying with a particular service. It is that, even while staying, one should know what one depends on and have the conditions to carry on the work when change becomes necessary. Export functions, readable records, clear termination procedures and the resetting of permissions can be those conditions. The convenience of a bot account must be assessed together with the user's ability to change the relationship.



〈Figure 28〉 Even after records and unfinished work are moved, the meaning of permissions and states must be confirmed again in the new service.

PART VI

Questions for Designing a Society of Bot Accounts



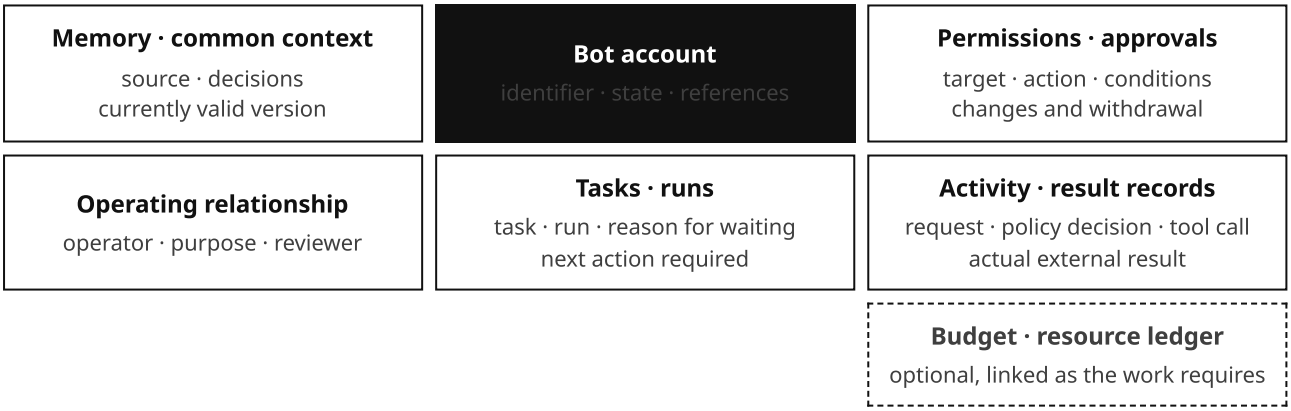
Chapter 21. Designing the Minimum Structure of a Bot Account

21.1 The Basic Structure: Identity, Operating Relationship, Permissions, Activity Record

Let us carry the discussion so far into an actual structure. When registering the publisher's research bot, what is needed first is a distinguishable identifier and an account state. To this are linked the operating manager, the purpose and the permitted scope of work. When a request arrives, task and run are distinguished and the results of external actions recorded. This structure is the starting point this book proposes, not the name of an established international standard.

“Minimum structure” does not mean forcing the same number of fields on every bot. **It means not omitting the questions that must be asked in actual work.** Who operates it, which permissions are valid, and which run did what must be answerable. For a simple function that takes no external action, the operating structure can be simpler too.

The management of authenticators and account information is also kept separate. Display name, description and role information may appear on work screens; secret keys and tokens are handled within an appropriate protection scheme. There is no need to replace the account structure with a scheme that stuffs every secret into the bot's long prompt. It is a design that divides storage and access according to the purpose for which information is used.



〈Figure 29〉 Identity, operator, permissions, tasks, memory and audit records are linked, but not every piece of data needs to be copied into the account record.

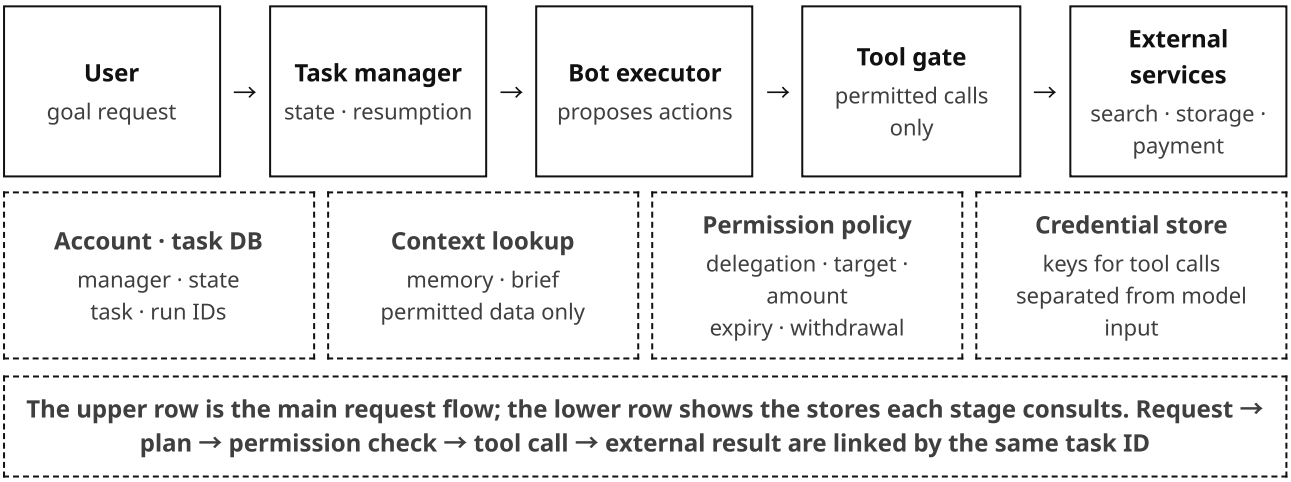
21.2 Connecting Memory, Task and Resource Management

An account can hold links referencing the memory store, task management, the budget ledger and audit records. Having linked them does not mean all the data must be copied into the account record. The systems that manage the originals can be kept distinct and made to reference the same identifiers. Only then can changes of different kinds, such as correcting memory and settling payments, be handled separately.

Consider the flow in which the research bot buys a paper. The user's request is registered as a task, a run proposes the purchase, and policy checks amount and target. After the necessary approval, external payment is made and the result is linked to the budget ledger and the work record. Once the material is stored, it can be referenced in memory with its source. If any one of these steps fails, it is not overwritten as complete; a state to be checked is left.

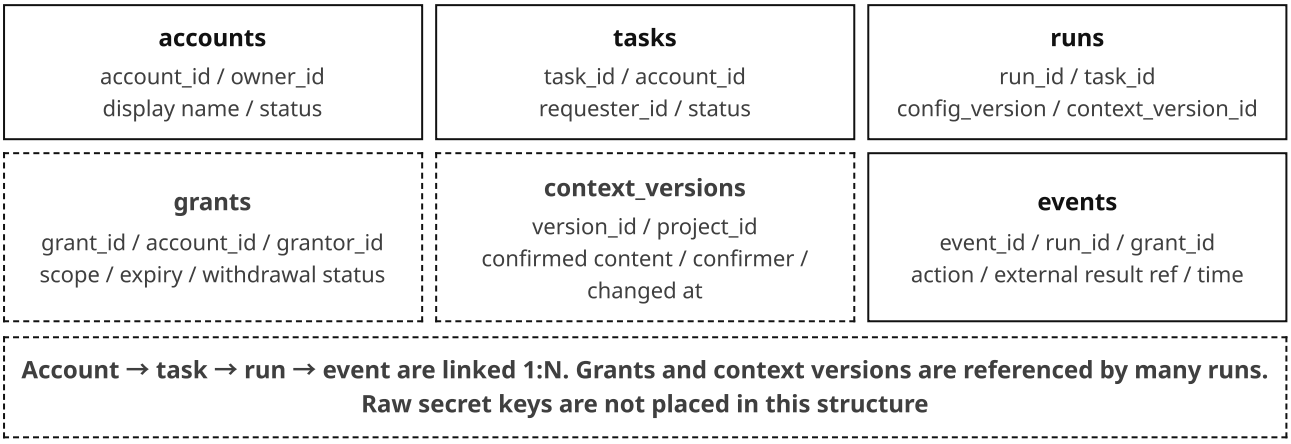
The common context supplies the purpose and criteria of this flow. The permission system confirms the allowed scope of action, and the audit record leaves the actual events. They are related, but no one function stands in for all the rest. In particular, the model's judgement that it has understood the goal well and the permission to pay must be evaluated separately.

Carried into a system configuration, this flow becomes a minimum arrangement that separates task management, bot execution, permission enforcement and external services. Initially it can consist of one app server, one database and one background executor. What matters is that **the model's proposal does not become execution authority by itself**, and that permission checks apply even to data access such as reading files.



〈Figure 30〉 The minimum arrangement from request to external action — each box need not be a separate server

Records are kept as linkable records rather than one giant log. Task creation, state changes and event records are stored consistently, together with per-organisation ID scopes and data access boundaries. Memory, outputs, purchases and budget, and approval history are extended as purpose requires.



〈Figure 31〉 The basic structure of linkable records — an explanatory data structure, not an operating schema

21.3 What Existing Standards Provide and What Still Needs Design

Existing technology supplies many parts. SCIM can be consulted for representing and managing account-related resources, the OAuth family for access and delegation, PROV for provenance relations, and A2A for communication between agents and the representation of tasks.[2][29][31][26] Even using each of these, the work of linking them to the organisation's working standards and approval procedures remains.

When combining standards, one must check whether the same word carries different meanings. The user of an account, the subject of a token, the agent in a provenance record and the person responsible for a task do not necessarily denote one object. Which identifiers to map between systems, and who may change that mapping, must be decided. Errors of linkage can arise even when each technology works correctly on its own.

The first implementation can therefore begin not by gathering every function but by checking whether one piece of work can be explained from end to end. Attach limited permissions and execution records to a bot that reads designated material and produces a draft. Later, when purchase or external contact is added, extend the approval, result confirmation and recovery procedures. An account structure that grows with the work in this way is what the design example at the end of Chapter 24 illustrates.

Chapter 22. If an Account Is Hijacked or Disappears

22.1 Bot Impersonation and Credential Leakage

Suppose an unfamiliar external transmission occurs under the research bot's name. The bot's authenticator may have leaked, or a bot running normally may have been influenced by instructions in malicious material. Both can lead to unpermitted actions, but the causes and responses differ. The first step is not to bundle every problem under the single phrase “account takeover.”

RFC 9700, which consolidates OAuth security best practice, addresses real threats such as token replay and the limiting of access rights.[38] The credentials a bot uses are likewise subject to protection and lifetime management. Prompt injection, by contrast, can affect the behaviour of a legitimate program without any credential leaking.[25] That identity verification succeeded does not allow the conclusion that the intent behind the action is safe.

To respond, it must be possible to confirm in which account, which run and which tool the abnormal behaviour appeared. If it is impersonation by a separate account with a similar name, verifying the counterpart matters; if the real account was compromised, existing permissions and runs must be examined. **The reason the earlier chapters distinguished identity from execution shows itself again in incident response.**

Credential leakage	Instructions mixed into external material
Starting point An issued secret key or token has passed to another party	**Starting point** A sentence in an external document is treated as a work instruction
Limit of successful authentication That a registered authenticator was used does not prove legitimate intent	**Limit of successful authentication** Even a properly authenticated bot can be influenced by external material
What to check Sessions, tokens, external connections and request logs; runs still alive after revocation	**What to check** The boundary between input material and instructions; tool calls, permissions and stored memory

Identity verification and action-level permission checks are designed together.

〈Figure 32〉 Distinguish a leaked authenticator from a legitimate bot influenced by instructions in external material, and check the records appropriate to the cause.

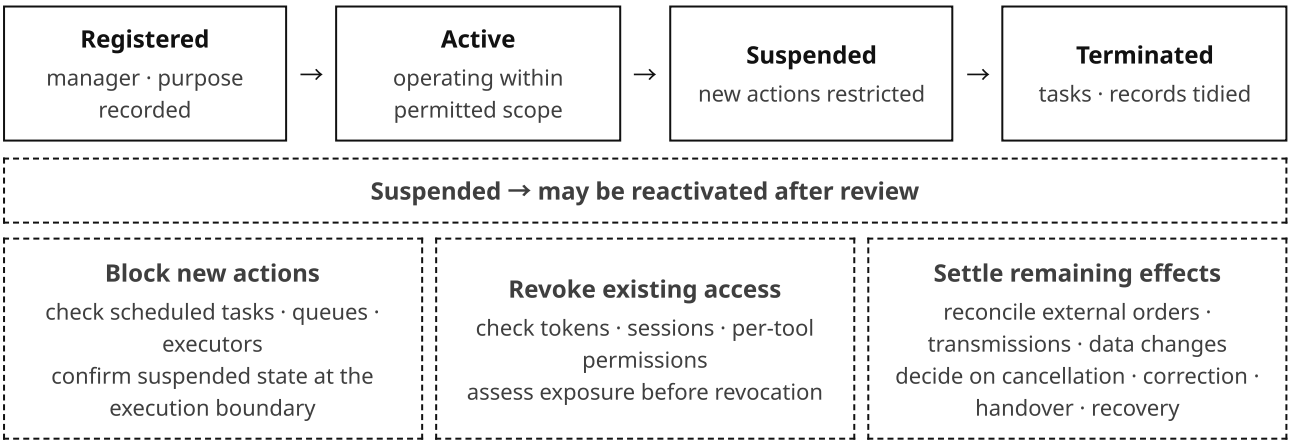
22.2 The Order of Revoking Permissions, Halting Activity and Recovering

When an incident is suspected, measures to stop the spread of harm must be weighed together with establishing the facts. In this book's publisher case, new tasks and external transmissions of the suspected account are halted, runs in progress and issued access rights are checked, and then which data was affected is investigated. This order is an example; actual response must be adjusted to the service structure and the urgency of the event.

It must not be assumed that changing the account password or key severs every connection already begun. Tokens and sessions, scheduled tasks and the permissions of subordinate tools must be checked together. Even after a run is stopped, if an external order or transmission has already completed, follow-up cancellation and notification may be needed. Recovery is broader than making the account screen openable again.

NIST SP 800-61 Rev. 3 links incident response to the organisation's risk management and sets out an approach to preparing detection, response and recovery.[39] The publisher, too, can avoid the state of searching for the location of records and the responsible person only after an incident. Setting points of contact, the authority to halt and the items to confirm on recovery in advance is practical operational preparation.

Changing account state, revoking external permissions and tidying unfinished work are different measures. **Suspension does not automatically reverse documents already sent or orders already completed, and termination differs from the immediate deletion of all records.** In implementation, account state, run cancellation, credential revocation and reconciliation of external results are each handled and their completion recorded. On recovery, not only permissions but memory, settings and pending tasks are checked, retention purposes and periods are set per data type, and necessary work is handed over.



〈Figure 33〉 Changes of account state and the three strands to check on suspension

22.3 Reconciling Privacy Protection with Evidence Preservation

Keeping every conversation and all personal information indefinitely on the ground of investigating an incident cannot be an appropriate default. At the same time, erasing all related records at once may make the extent of harm hard to establish. The scope of necessary data, the basis for retention, who may access it and for how long must be set. The destruction principle of Korea's Personal Information Protection Act and its statutory retention exceptions are a starting point for examining this distinction.[24] The memory of a recovered bot must be checked too. If wrong instructions or data were stored during the compromise, the same errors may recur even after new credentials are issued. Set the criteria for returning to a normal state, and review the changed permissions, memory and tasks in progress. This process shows the difference between reviving an account and making operation trustworthy again.

One can also imagine a service disappearing entirely. Unfinished tasks and necessary outputs must be able to pass to another person. The persistence of an account does not depend on one program running forever. A structure in which, on interruption, tasks and records can be confirmed and properly handed over is what gives the relationship its resilience.

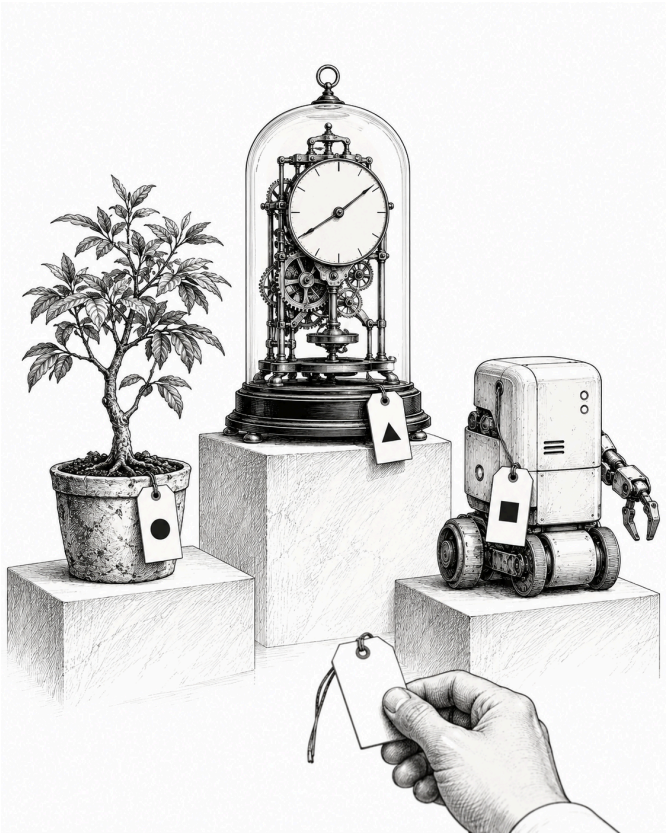
Chapter 23. Does an Account Create a Person?

23.1 Technical Subject, Social Role and Legal Personhood

When a bot works for a long time with a name and a memory, people may come to refer to it as a counterpart. “I left it to the research bot” or “the editing bot missed it” sound natural. But everyday address, identification within a system and legal status are different questions. Calling them by the same word does not merge the layers.

A technical subject can be understood as the unit that links requests and records. A social role concerns the functions people expect of and entrust to that counterpart. Legal personhood is the institutional status of being recognised as a subject of rights and duties. The Korean Civil Act deals with the legal capacity of natural persons and of juridical persons in separate articles.[12] Such distinctions remind us that legal status cannot be derived from the issue of an account alone.

The fact that W3C DID Core allows subjects other than people to be identified must be read in the same light.[10] That a digital object can be distinguished describes the scope of an identification scheme. It is not evidence that the object has consciousness or moral status. **That a technical specification uses the word “subject” does not bring the human concept of personhood along with it.**



〈Figure 34〉 Numbers and accounts are devices for identifying an object. They alone do not prove consciousness or legal personhood.

23.2 Questions around AI Ownership, Rights and Responsibility

If an organisation operates bot accounts, it must decide who controls the software, the stored data, the generated results and the access rights. These matters may involve different contracts and legal relationships. The phrase “owning a bot” is too broad unless it states what authority over what is meant. Control of user data, copyright in the program and the company's operating authority must be distinguished.

Whether to grant some AI a separate legal status in future can be discussed as a matter of institutional choice. But to examine its necessity and effect one must ask whose interests and responsibilities actually change. A structure that gives a bot responsibility in name only while blurring the responsibility of the operating organisation may make harm harder to put right. This is a concern for evaluating possible institutional designs, not an assertion about the state of every present system.

At the same time, the possibility of consciousness can be neither excluded nor confirmed by the presence of an account. The 2023 report by Butlin and colleagues proposed an approach that derives indicators from theories of consciousness and examines AI systems against them.[40] That report's assessment of systems at the time must not be extended into a final conclusion about all later AI. Success in account management and the existence of subjective experience call for different kinds of evidence.

23.3 The Boundary between Technical Fact and Institutional Choice

Suspending the philosophical question does not mean operating rules must be postponed too. If a bot reads external material and performs work that affects people, permissions, records and channels for explanation and correction are needed. This demand can be justified without assuming the bot is conscious, because the relationship between the people who suffer harm and the organisation that operates it already exists.

Conversely, one cannot claim that good technical management alone has resolved every future moral question. An account must make clear both what can be recorded and what records alone cannot answer. Confusion lessens when the system that manages observable behaviour and the debate that judges the nature of a being each keep to their own scope.

The claim of this book, then, is not to grant bots the human resident registration as it stands. It is to re-ask, for the behaviour of software, the questions of identification and relationship management developed in human society. The fact that an account has been created does not define every status of the counterpart. It does offer an opportunity to explain more clearly what relationship we have decided to have with it.

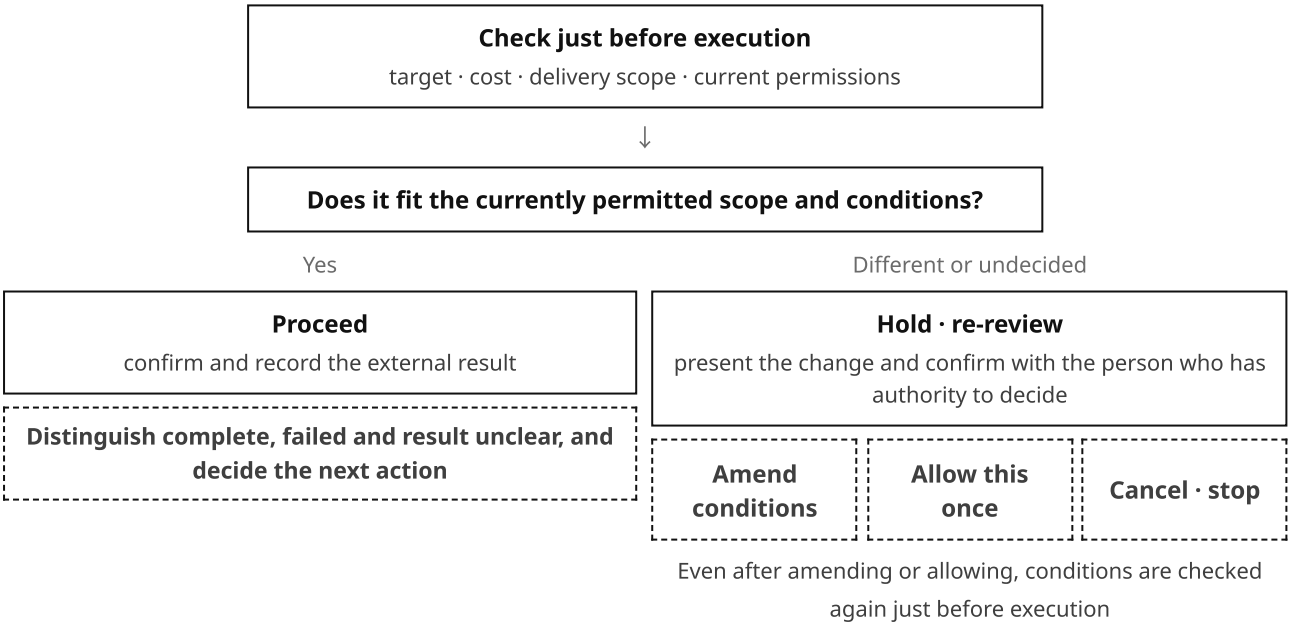
Chapter 24. What Should We Entrust to AI?

24.1 Conditions under Which People Can Decide and Review

To entrust work to an AI is to depend on another process for part of one's judgement. If every judgement is repeated personally each time, the benefit of delegation shrinks. But there is no need to accept a state in which results cannot be reviewed at all. The real choice is distinguishing the scope that can be entrusted from the conditions under which judgement must be made again.

The publisher can entrust material collection and draft preparation broadly while setting separate conditions for new spending or the external delivery of unpublished manuscripts. For a person to be able to review, the necessary facts must be presented at the right moment. Providing a long conversation history after execution differs from showing the target and impact before it.

Even with the authority to review, lacking time or being unable to understand the conditions can reduce approval to a formality. This book proposes presenting important choices linked to the concrete change involved, and managing repeatedly approved scopes through clear policy. Keeping human judgement in the loop is not a matter of adding one more button; it is a matter of adjusting the pace of work and the provision of information together.



〈Figure 35〉 Proceed within the permitted scope; review again when target, cost or impact changes. People must be able to stop or amend the conditions.

24.2 The Option to Stop, Correct and Object

Those affected by a bot's work may not be only its direct users. A person who receives an automatically sent message, a person who appears in the material, a person subjected to automated assessment are all part of the relationship. The user's consent therefore cannot be taken to settle every interest involved. Channels for explanation and correction must be provided according to the external impact.

Article 37-2 of Korea's Personal Information Protection Act provides for refusal and requests for explanation regarding fully automated decisions made by processing personal information. The right to refuse carries the requirement of a significant effect on rights or duties and statutory exceptions, and automated dispositions by administrative agencies are excluded from the article.[41] This must not be described as a general right to refuse every answer of every bot.

The operating principle this book proposes is distinct from the scope of any individual law: that work can be stopped, wrong premises corrected and insufficiently explained results re-examined by a person. In particular, these options must not vanish because of an account suspension or a service transfer. Continuous delegation requires a continuous channel for changing the relationship.

24.3 Possible Futures and the Principles We Set

The first future one can imagine is a structure in which a few platforms manage bots' identity, memory and work together. It may offer an integrated experience, but the conditions for changing services and moving data may become important. The second is a structure in which many services connect through standards and contracts. The range of choice may widen, but the cost of aligning different policies and states remains.

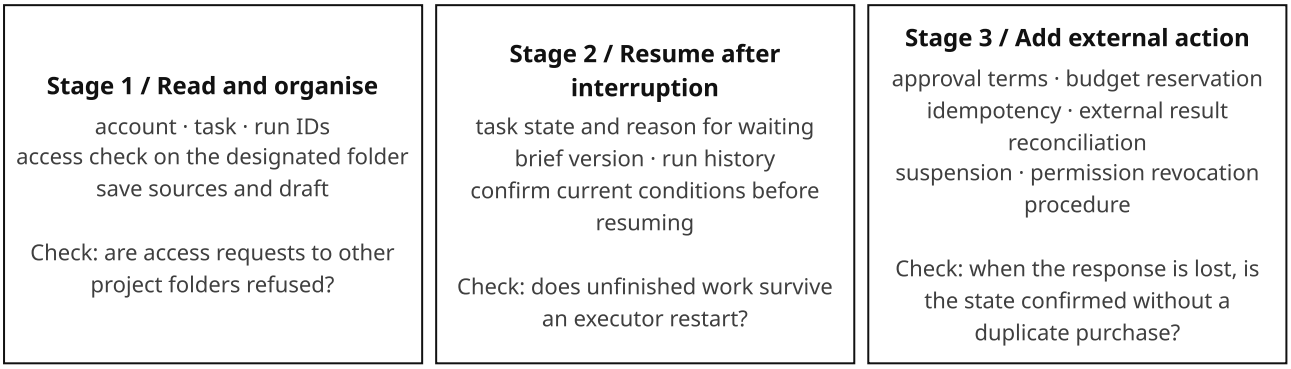
Both are possible scenarios; they are not predictions of adoption rates or timing.

The third is a structure in which strictly limited bot accounts operate only within particular tasks in each organisation. With few external links the scope may be easier to explain, but connecting a user's several tasks into one flow may be constrained. In practice these approaches may mix. There is no basis for declaring any one future the inevitable conclusion of technical progress.

This book proposes criteria for evaluating each structure. Can the agent of an action and the delegation relationship be confirmed? Can only the necessary scope be allowed and changes reflected? Can results and sources be reviewed and wrongs put right? Can the necessary working relationships be settled even when a service is changed or ended? If bot accounts have value, it lies in making these questions answerable in concrete terms.

24.4 A First Implementation Connects One Task End to End

If the diagrams in this book were carried into practice, it would be better to start not by gathering every function but by connecting one task end to end. An example goal is to read material from a designated folder and produce a draft reference list with sources. The work is divided into three stages, each with a question to check.



〈Figure 36〉 Three stages of a first implementation — example goal: read material from a designated folder and produce a draft reference list with sources

To Grant an Account Is to Design a Relationship

Let us return to the opening question. What do a resident registration number, an account number and an internet account each do? We use them often, but we do not think each time about the relationships behind the numbers and names. They can be used in daily life because procedures operate by which records continue, access is permitted and wrong information is corrected.

Thinking about bot accounts is re-examining these procedures for the behaviour of software. **More important than the moment of naming a bot is the time after the work has been entrusted.** How far it may proceed while the user is away, what must be rechecked when conditions change, and who can put things right when an error occurs determine the character of the relationship.

This book has not claimed that an account makes an AI a person. It has proposed instead that we not confuse the technically distinguishable subject, the unit responsible for work and the party whose legal responsibility is to be judged. Only with that distinction can we examine the possibilities of AI without losing sight of the relationships of the people and organisations affected now.

Memory helps carry on the work that is needed, and the common context aligns the direction of many participants. Permissions define the permitted actions, and task state reveals what remains. Provenance and audit records help confirm results. When these functions are connected, a user can explain the statement “I work continuously with a particular bot” as an actual operating structure.

Above all, the relationship must be changeable. One must be able to entrust new work, correct wrong memories, reduce permissions and, if necessary, stop or move to another service. The more useful a bot becomes, the better the user must be able to understand the terms of the relationship. Gaining convenience is no reason to lose the ability to correct and to object.

What we have to design is not only a society in which accounts multiply. **It is a society in which the work entrusted to AI can be explained within human judgement and institutions.** Bot accounts can be a starting point for asking about that relationship concretely.

Appendix

Appendix A. Key Terms

An identifier is a value that distinguishes an object within a defined scope. The same string may point to different objects under different issuing systems. A display name is an expression people read and use; it may serve as an identifier but is not always unique. Authentication is the process of confirming the authenticator linked to a claimed identity. Authorisation is the process of deciding whether to permit a requested action or access. Successful authentication alone does not permit every action.

An account, in this book, means the unit of management that links a counterpart's identity and information, access relationships and operating state. Definitions of “account” in individual standards and institutions may differ. A bot account applies this concept to the continuous working relationship of software.

A model is the core component that processes input and generates output. A bot is used to mean a software configuration that combines instructions, tools and state so as to perform work. The term agent focuses on the function of choosing the next action and continuing work on receiving results.

A run is the unit of an individual event in which a program operated under a certain configuration and conditions. A task is the unit of work managed to achieve the user's purpose and may span several runs. Account, task and run are linked but are not the same concept.

Delegation is the relationship in which one party entrusts another with carrying out work within a defined scope. The actual granting of permissions is implemented through the rules of services and organisations. Whether the person requesting work is able to grant the relevant permission must also be confirmed. The common context is the goals, material, constraints, decisions and current state managed so that participants in a task reference them together. It is an operating concept used in this book and does not mean unlimited information sharing or a single established international standard.

A provenance record explains through which activities and agents a piece of data was produced. An audit record is kept to confirm later a particular event and its conditions. The existence of a record and the truth of its content are examined separately.

Idempotency here means the property of handling a retried request of the same intent so that no unintended duplicate effect occurs. The actual scope and duration of the guarantee and the handling of errors must be checked for each service.

Governance is the operating and decision-making structure that sets who decides and reviews by what criteria, and who puts problems right. Technical attribution links an action to an identity and a run; legal responsibility is a matter judged by the applicable law and the facts.

Appendix B. References and Notes

Sources were checked on 28 October 2026. Statutes are cited in the versions in force as indicated below; technical specifications and studies in the versions stated. Reference numbers are shared across the main text and the appendices.

[1] NIST. Digital Identity Guidelines, SP 800-63-4 (2025), Appendix A Glossary, “digital identity”, “identifier”.
Direct basis for uniqueness within a given context. The staff-number example was constructed to illustrate the definition; the definition is not applied unchanged to identity concepts in every field.

[2] Hunt, P., et al. System for Cross-domain Identity Management: Core Schema. RFC 7643 (2015), §§3.1, 4.1.1.
A standard that distinguishes id, userName and displayName. The stability and non-reassignment requirements concern SCIM resources and do not mean that every service's login name is immutable.

[3] ISO. ISO 13616-1:2020, Financial services — International bank account number (IBAN) — Part 1: Structure of the IBAN, §3.1.
Publicly available term definition. Used to distinguish account number from account; no conclusions about any country's deposit contracts or ownership are drawn from it.

[4] NIST. Digital Identity Guidelines, SP 800-63-4 (2025), §2 Digital Identity Model.
Direct basis for the relationship between subscriber account, verifier and authenticator. The subject in this document is a person; extending the distinction to bot account design is the author's application.

[5] NIST. Digital Identity Guidelines: Authentication and Authenticator Management, SP 800-63B-4 (2025), §3.1.1 Passwords.
Basis for treating a password as a secret used in authentication. The text does not describe the document's detailed password policy or recommend a particular method.

[6] Hardt, D., ed. The OAuth 2.0 Authorization Framework. RFC 6749 (2012), §2.2 Client Identifier.
Basis for the rule that client_id alone must not authenticate a client. An OAuth client is an application, not a user's login name. Later security guidance has updated parts of the specification; only the conceptual distinction is used here.

[7] Hu, V. C., et al. Guide to Attribute Based Access Control (ABAC) Definition and Considerations. NIST SP 800-162 (2014; updated 2019).
Describes policy evaluation based on subject, object, operation and environment. The company example separating read and edit rights is a fictional application.

[8] NIST. Digital Identity Guidelines: Identity Proofing and Enrollment, SP 800-63A-4 (2025), §5 Subscriber Accounts.
Basis for pseudonymous accounts, roles and multiple accounts, changes of information, suspension and termination. The same duties do not automatically arise for services outside the scope of the guideline.

[9] Searle, J. R. (2015). Status functions and institutional facts: reply to Hindriks and Guala. Journal of Institutional Economics, 11(3), 507–514, esp. p. 508. DOI: 10.1017/S1744137414000629.
A philosophical discussion of status functions, rights and duties. Not used as a technical definition of digital accounts or empirical research on AI personhood. “A place within a system” is the author's interpretation.

[10] W3C. Decentralized Identifiers (DIDs) v1.0. W3C Recommendation, 19 July 2022, Abstract and §2 Terminology, “DID subject”.
Supports the point that identified subjects are not limited to people. DID is not proposed as a required technology for bot accounts, and no consciousness or legal personhood is inferred from identifiability.

[11] Republic of Korea. Resident Registration Act. In force 22 July 2025. Articles 7, 7-2, 21.
Used to distinguish individual and household records, the unique number, and objections to certain dispositions.

[12] Republic of Korea. Civil Act. In force 17 March 2026. Articles 3 and 34.
Basis for the legal capacity of natural and juridical persons. The whole legal status of AI is not settled by these two articles.

[13] United Nations Statistics Division. United Nations Legal Identity Agenda. Background and UN Operational Definition of Legal Identity.
Confirms the scope of international discussion of recognition before the law, birth registration and legal identity.

[14] UNICEF Data. Birth registration.
Material on the relation between birth registration and access to essential services. Changing registration figures are not reproduced in the text.

[15] McLeay, M., Radia, A., & Thomas, R. (2014). Money in the modern economy: an introduction. Bank of England Quarterly Bulletin, 2014 Q1.
Used to explain the economic nature of bank deposits. UK material is not substituted for the legal interpretation of Korean deposit contracts.

[16] Republic of Korea. Act on Real Name Financial Transactions and Confidentiality. In force 1 April 2025. Article 3.
Basis for real-name transactions and the presumption of ownership of financial assets in certain accounts.

[17] NIST. (2025). Digital Identity Guidelines: Federation and Assertions. SP 800-63C-4, §1 Introduction.
Describes the structure of federated authentication. Not used to mean that all per-service data and permissions are merged.

[18] W3C. (2021). Web Authentication: An API for accessing Public Key Credentials, Level 2. Recommendation, 8 April 2021.
An example of authentication using public-key credentials. Not used as a description of current product support.

[19] Yao, S., et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023; arXiv:2210.03629v3.
Research linking reasoning with external actions. Experimental results are not generalised to all real work.

[20] SPIFFE Project. SPIFFE Concepts; SPIFFE Overview. Official technical documentation.
Consulted for workload identity, multiple running instances and the distinction of attestation documents. Not a standard for bot accounts as a whole.

[21] NIST Joint Task Force. (2020). Security and Privacy Controls for Information Systems and Organizations. SP 800-53 Rev. 5. AC-2, AC-6, AU-9.
Referenced for account management, least privilege and protection of audit information. The Release 5.2.0 notice on the public page was also checked.

[22] Park, J. S., et al. (2023). Generative Agents: Interactive Simulacra of Human Behavior. UIST 2023; arXiv:2304.03442v2.
Consulted for the memory, planning and reflection structure in a simulated environment. Not used as evidence of human-like memory experience.

[23] Packer, C., et al. (2024). MemGPT: Towards LLMs as Operating Systems. arXiv:2310.08560v2; first released 2023.
Research on managing the input context and an external memory tier. Not material that guarantees the completeness or accuracy of actual memory.

[24] Republic of Korea. Personal Information Protection Act. In force 11 September 2026. Article 21.
Distinguishes the principle of destruction of personal information from retention exceptions under other laws.

[25] OWASP Gen AI Security Project. LLM01:2025 Prompt Injection.
Describes the risk that external material influences model behaviour. The proposed controls are not claimed to guarantee complete defence.

[26] A2A Project. Agent2Agent Protocol Specification, version 1.0.1. §§3.4, 4.1, 4.4.
Consulted for task and context identifiers, the input-required state and the Agent Card. The communication format is distinguished from guarantees of work quality.

[27] Temporal Technologies. Temporal Workflow Execution overview. Official technical documentation.
An example of a structure that preserves task state and resumes execution. Not used to mean unconditional exactly-once execution of external side effects.

[28] Stripe. Idempotent requests. API reference.
An example of handling duplicate effects of retried requests. Specific guarantees and error handling must be checked against the API version and conditions.

[29] Jones, M., et al. (2020). OAuth 2.0 Token Exchange. RFC 8693, §§1.1, 4.1.
Technical basis for distinguishing delegation, impersonation and the acting party. Protocol-level impersonation is not translated directly into unlawful posing.

[30] Lodderstedt, T., Richer, J., & Campbell, B. (2023). OAuth 2.0 Rich Authorization Requests. RFC 9396, §§1, 2.
Consulted for fine-grained authorisation details and the payment example. Expression of conditions is distinguished from actual enforcement.

[31] Moreau, L., & Missier, P., eds. (2013). PROV-DM: The PROV Data Model. W3C Recommendation, 30 April 2013. §§2, 5.3.
Basis for expressing provenance relations among entities, activities and agents. Not used to adjudicate legal responsibility or guarantee the truth of records.

[32] United Nations. (2005). Convention on the Use of Electronic Communications in International Contracts. Article 12; UNCITRAL explanatory note.
Explains the relation between automated message systems and contracts. The convention's scope and the validity of individual contracts are separate matters.

[33] UNCITRAL. (2024). Model Law on Automated Contracting. Adopted 11 July 2024.
A legislative model for automated contracting. Not presented as a treaty or a uniform law directly applicable in each country.

[34] NIST. (2023). Artificial Intelligence Risk Management Framework, AI RMF 1.0. NIST AI 100-1. GOVERN 1.6, 2.1.
Consulted for the inventory of AI systems and roles and responsibilities. Version 1.0 is cited; it is a framework for voluntary use.

[35] NIST. (2024). Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile. NIST AI 600-1, §2.7.
Basis for reviewing risks in human–AI interaction. It does not diagnose psychological effects in any particular user.

[36] European Union. Regulation (EU) 2016/679, General Data Protection Regulation. Article 20.
Describes the conditional right to data portability. Not extended into a general right to move a bot's whole state, model or organisational data.

[37] Iannella, R., & Villata, S., eds. (2018). ODRL Information Model 2.2. W3C Recommendation, 15 February 2018.
A model for expressing permissions, prohibitions and duties of use. Actual enforcement and legal effect are separate.

[38] Lodderstedt, T., Bradley, J., Labunets, A., & Fett, D. (2025). Best Current Practice for OAuth 2.0 Security. RFC 9700, §§2.2, 2.3.
Consulted for preventing token replay and limiting access token privileges, alongside the original OAuth 2.0 specification.

[39] Nelson, A., Rekhi, S., Souppaya, M., & Scarfone, K. (2025). Incident Response Recommendations and Considerations for Cybersecurity Risk Management: A CSF 2.0 Community Profile. NIST SP 800-61 Rev. 3.
Basis for linking incident detection, response and recovery to organisational risk management.

[40] Butlin, P., et al. (2023). Consciousness in Artificial Intelligence: Insights from the Science of Consciousness. arXiv:2308.08708v3.
Consulted for the indicator approach based on theories of consciousness. The 2023 assessment is not used as a conclusion about all AI after 2026.

[41] Republic of Korea. Personal Information Protection Act. In force 11 September 2026. Article 37-2.
Describes rights regarding fully automated decisions, distinguishing the requirements and exceptions of the right to refuse and the dispositions excluded.

[42] Rose, S., Borchert, O., Mitchell, S., & Connelly, S. (2020). Zero Trust Architecture. NIST SP 800-207.
Consulted for the approach that does not grant trust by location or ownership alone and for the distinction between authentication and authorisation.

[43] Ha, Woo Sung. (2026). The Knowledge–Action Chain (KAC): An AX Execution Ontology for Turning Validated Knowledge into Action, Outcome, and Learning. Unpublished manuscript, 23 June 2026. Korean and English editions: github.com/sopia19910/Knowledge-Action-Chain.
Cited for the nine slots of the Common Context Structure (CCS), the Domain Context Integrity (DCI) gate, the SkillRuntime, the distinction between output and outcome, and the cycle of review, feedback and context update. An execution model proposed by the author; it states that the verification criteria and coefficients of each segment are operating variables to be tuned with real data. The interpretation linking it to bot accounts in Sections 10.4 and 15.4 is this book's own.